

# Physics Publications with Major Personal Contributions

Oliver Gutsche

May 05, 2026

1. Observation of the rare decay  $\eta \rightarrow \mu^+ \mu^- e^+ e^-$

Hayrapetyan, Aram et al.

[arXiv:2605.00615](#) | 2026-05-01

2. Search for light charged Higgs bosons decaying to charm and strange quarks in  $t\bar{t}$  events in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2604.27795](#) | 2026-05-01

3. Improved results on Higgs boson pair production in the  $4b$  final state

Hayrapetyan, Aram et al.

[arXiv:2604.27044](#) | 2026-04-29

4. Search for electroweakinos in compressed-spectrum scenarios with low-momentum isolated tracks in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2604.25604](#) | 2026-04-28

5. Simultaneous measurements of  $N$ -subjettiness observables in jets from gluons and light-flavour quarks, and in decays of boosted  $W$  bosons and top quarks

Hayrapetyan, Aram et al.

[arXiv:2604.25538](#) | 2026-04-28

6. Measurement of the  $Z \rightarrow \mu^+ \mu^-$  angular coefficients in  $pp$  collisions at  $\sqrt{s} = 13$  TeV as functions of transverse momentum and rapidity

Gevorgyan, Arzunik et al.

[arXiv:2604.25678](#) | 2026-04-28

7. Search for associated production of a Higgs boson and two vector bosons via vector boson scattering at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2604.24531](#) | 2026-04-27

8. Search for dark matter produced in association with a dark Higgs boson decaying into a bottom quark-antiquark pair in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2604.20999](#) | 2026-04-22

9. Search for the single production of vector-like quarks decaying into a  $W$  boson and a  $b$  quark using single-lepton final states in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2604.17564](#) | 2026-04-19

10. Highly boosted dielectron identification in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2604.13320](#) | 2026-04-14

11. Search for heavy resonances decaying into four-lepton final states via light bosons in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2604.14236](#) | 2026-04-14

12. A search for microscopic black holes, string balls, and sphalerons in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2604.10732](#) | 2026-04-12

13. Search for a new heavy resonance decaying to a top quark and a neutral scalar boson in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2604.10729](#) | 2026-04-12

14. Particle transformers for identifying Lorentz-boosted Higgs bosons decaying to a pair of W bosons

Hayrapetyan, Aram et al.

[arXiv:2604.09809](#) | 2026-04-10

15. Observation of nuclear suppression in coherent  $\Upsilon(1S)$  photoproduction off heavy nuclei at the LHC

Belyaev, Andrey et al.

[arXiv:2604.05814](#) | 2026-04-07

16. Search for soft unclustered energy patterns produced in association with a W or Z boson in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2604.05996](#) | 2026-04-07

17. Evidence of  $ZZ\gamma$  production and observation of  $4\ell\gamma$  in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2604.02594](#) | 2026-04-03

18. Search for Higgs boson pair production in the  $b\bar{b}WW$  decay channel with two leptons in the final state using proton-proton collision data at  $\sqrt{s} = 13.6$  TeV

Gevorgyan, Arzunik et al.

[arXiv:2604.02127](#) | 2026-04-02

19. Measurement of dijet angular distributions and search for beyond the standard model physics in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Gevorgyan, Arzunik et al.

[arXiv:2603.25458](#) | 2026-03-26

20. Search for new particles decaying into top quark-antiquark pairs in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Gevorgyan, Arzunik et al.

[arXiv:2603.23454](#) | 2026-03-24

21. Search for low-mass vector and scalar resonances decaying into a quark-antiquark pair in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2603.21965](#) | 2026-03-23

22. Search for Higgs boson production at high transverse momentum in the WW decay channel in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2603.22233](#) | 2026-03-23

23. Measurement of the jet mass in hadronic decays of boosted W bosons at 13 TeV and extraction of the W boson mass

Hayrapetyan, Aram et al.

[arXiv:2603.19963](#) | 2026-03-20

24. Jet peak shapes based on two-particle angular correlations in lead-lead collisions at  $\sqrt{s_{NN}} = 5.02$  TeV

Belyaev, Andrey et al.

[arXiv:2603.14385](#) | 2026-03-15

25. Measurement of the t-channel single top quark cross section in proton-proton collisions at  $\sqrt{s} = 5.02$  TeV

Hayrapetyan, Aram et al.

[arXiv:2603.13592](#) | 2026-03-13

26. Search for Z' bosons decaying into charginos in final states with two oppositely charged leptons and missing transverse momentum in pp collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2603.11035](#) | 2026-03-11

27. Measurement of angular correlations inside jets induced by gluon polarization in proton-proton collisions at  $\sqrt{s} = 13.6$  TeV

Hayrapetyan, Aram et al.

[arXiv:2603.03689](#) | 2026-03-04

28. Search for a narrow resonance with a mass between 10 and 70 GeV decaying to a pair of photons in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2603.03250](#) | 2026-03-03

29. Combination of ATLAS and CMS searches for Higgs boson pair production at  $\sqrt{s} = 13$  TeV

Aad, Georges et al.

[arXiv:2602.23991](#) | 2026-02-27

30. System-size dependence of charged-particle suppression in ultrarelativistic nucleus-nucleus collisions

Belyaev, Andrey et al.

[arXiv:2602.21325](#) | 2026-02-24

31. Search for pair production of heavy resonances in final states with a photon and large-radius jets in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2602.20477](#) | 2026-02-24

32. Observation of the jet diffusion wake using dijets in heavy ion collisions

Hayrapetyan, Aram et al.

[arXiv:2602.19431](#) | 2026-02-23

33. Search for a new resonance decaying to a Higgs boson and a scalar boson in events with two b jets and two Z bosons in proton-proton collisions at  $\sqrt{s} = 13.6$  TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18223](#) | 2026-02-20

34. Combined measurements and interpretations of Higgs boson production and decay in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18611](#) | 2026-02-20

35. First measurement of jet axis decorrelation with photon-tagged jets in pp and PbPb collisions at 5.02 TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18279](#) | 2026-02-20

36. Measurement of event shape variables using charged particles inside jets in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2602.17509](#) | 2026-02-19

37. Influence of strangeness on the anisotropic flow of prompt  $D^{\pm}_s$  mesons in PbPb collisions at  $\sqrt{s_{NN}} = 5.02$  TeV

Belyaev, Andrey et al.

[arXiv:2602.14221](#) | 2026-02-15

38. High-level hadronic tau lepton triggers of the CMS experiment in proton-proton collisions at  $\sqrt{s} = 13.6$  TeV

Hayrapetyan, Aram et al.

[JINST 21 \(2026\) P04002](#) | [arXiv:2602.11359](#) | [DOI:10.1088/1748-0221/21/04/P04002](#) | 2026-02-11

39. Measurement of B meson production fraction ratios in proton-proton collisions at  $\sqrt{s} = 13$  TeV using open-charm and charmonium decays

Hayrapetyan, Aram et al.

[arXiv:2602.10270](#) | 2026-02-10

40. Probing early parton emissions in heavy ion collisions using the Lund jet plane

Belyaev, Andrey et al.

[arXiv:2602.09271](#) | 2026-02-09

41. Observation of  $\Upsilon(1S) + Z$  associated production and measurement of the effective double-parton scattering cross section in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2602.02770](#) | 2026-02-02

42. Observation of a family of all-charm tetraquarks

Hayrapetyan, Aram et al.

[arXiv:2602.02252](#) | 2026-02-02

43. Search for heavy scalar resonances decaying to Lorentz-boosted Higgs and Higgs-like bosons in the  $b\bar{b}4q$  final state at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2602.00273](#) | 2026-01-30

44. Measurement and effective field theory interpretation of the photon-fusion production cross section of a pair of W bosons in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2601.21574](#) | 2026-01-29

45. Measurement of the  $\Upsilon(1S)$ ,  $\Upsilon(2S)$ , and  $\Upsilon(3S)$  differential cross sections in pp collisions at  $\sqrt{s} = 13.6$  TeV

Hayrapetyan, Aram et al.

[arXiv:2601.20023](#) | 2026-01-27

46. Search for heavy resonances decaying into two Higgs bosons in the  $b\bar{b}\tau^+\tau^-$  final state in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2601.20011](#) | 2026-01-27

47. Search for heavy long-lived charged particles with level-1 trigger scouting data from proton-proton collisions at  $\sqrt{s} = 13.6$  TeV

Hayrapetyan, Aram et al.

[arXiv:2601.20063](#) | 2026-01-27

48. Strategy and performance of the CMS long-lived particle trigger program in proton-proton collisions at  $\sqrt{s} = 13.6$  TeV

Hayrapetyan, Aram et al.

[arXiv:2601.17544](#) | 2026-01-24

49. Search for the pair production of long-lived supersymmetric partners of the tau lepton in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2601.17576](#) | 2026-01-24

50. Full event interpretation with machine-learning-based particle-flow reconstruction in the CMS detector

Hayrapetyan, Aram et al.

[arXiv:2601.17554](#) | 2026-01-24

51. Measurement of the  $Z\gamma$  production cross section and search for anomalous neutral triple gauge couplings in pp collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2601.14102](#) | 2026-01-20

52. Combination of searches for heavy vector boson resonances in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2601.12583](#) | 2026-01-18

53. Search for dark matter produced in association with a Higgs boson decaying to bottom quarks in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2601.11330](#) | 2026-01-16

54. Search for a boosted Higgs boson decaying to bottom quark pairs in association with a W or Z boson in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2601.05362](#) | 2026-01-08

55. Measurement of the Higgs boson total decay width using the  $H \rightarrow WW \rightarrow e\nu\mu\nu$  decay channel in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2601.05168](#) | 2026-01-08

56. Search for exotic Higgs boson decays  $H \rightarrow \mathcal{A}\mathcal{A}$  with  $\mathcal{A}\mathcal{A} \rightarrow \gamma\gamma$  in events with a semi-merged topology in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2601.00183](#) | 2026-01-01

57. Search for charged Higgs bosons decaying into top and bottom quarks in lepton+jets final states in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[arXiv:2512.24471](#) | 2025-12-30

58. Machine-learning techniques for model-independent searches in dijet final states

Hayrapetyan, Aram et al.

[arXiv:2512.20395](#) | [DOI:10.5281/zenodo.16656501](#) | 2025-12-23

59. Characterization of the quantum state of top quark pairs produced in proton-proton collisions at  $\sqrt{s}=13\text{TeV}$  using the beam and helicity bases

Hayrapetyan, Aram et al.

*Phys.Rev.D* 113 (2026) L051101 | [arXiv:2512.17557](#) | [DOI:10.1103/hkgw-dztk](#) | 2025-12-19

60. Measurement of  $Z\gamma$  production in proton-proton collisions at  $\sqrt{s} = 13.6 \text{ TeV}$  and constraints on neutral triple gauge couplings

Hayrapetyan, Aram et al.

[arXiv:2512.08582](#) | 2025-12-09

61. Measurements of electroweak production of a photon in association with two jets in proton-proton collisions at  $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[arXiv:2512.00502](#) | 2025-11-29

62. Search for light pseudoscalar bosons, pair-produced in Higgs boson decays in the four-electron final state in proton-proton collisions at  $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[arXiv:2511.19563](#) | 2025-11-24

63. Search for new physics in the final state with a single photon and large missing transverse momentum in proton-proton collisions at  $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

*Phys.Rev.D* 113 (2026) 072020 | [arXiv:2511.17310](#) | [DOI:10.1103/f9r8-styj](#) | 2025-11-21

64. Search for Higgsinos in final states with low-momentum lepton-track pairs at 13 TeV

Hayrapetyan, Aram et al.

[arXiv:2511.16394](#) | 2025-11-20

65. Search for heavy  $H\gamma$  and  $Z\gamma$  resonances with a bottom quark-antiquark pair in the final state in proton-proton collisions at  $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[arXiv:2511.14583](#) | 2025-11-18

66. Search for low-mass hidden-valley dark showers with non-prompt muon pairs in proton-proton collisions at  $\sqrt{s}=13 \text{ TeV}$

Hayrapetyan, Aram et al.

*JHEP* 03 (2026) 189 | [arXiv:2511.11888](#) | [DOI:10.1007/JHEP03\(2026\)189](#) | 2025-11-14

67. Search for a new neutral gauge boson produced in association with one or two b jets and decaying into a pair of muons in proton-proton collisions at  $\sqrt{s} = 13 \text{ TeV}$

Chekhovsky, Vladimir et al.

[arXiv:2511.11853](#) | 2025-11-14

68. Search for long-lived particles using displaced vertices with low-momentum tracks in proton-proton collisions at  $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[arXiv:2511.08212](#) | 2025-11-11

69. Identification of tau leptons using a convolutional neural network with domain adaptation

Hayrapetyan, Aram et al.

*JINST* 20 (2025) P12032 | [arXiv:2511.05468](#) | [DOI:10.1088/1748-0221/20/12/P12032](#) | 2025-11-07

70. Jet fragmentation function and groomed substructure of bottom quark jets in proton-proton collisions at 5.02 TeV  
Chekhovsky, Vladimir et al.  
[JHEP 04 \(2026\) 147](#) | [arXiv:2511.10666](#) | [DOI:10.1007/JHEP04\(2026\)147](#) | 2025-11-06
71. Inclusive and differential measurements of the  $t\bar{t}\gamma$  cross section and the  $t\bar{t}\gamma / t\bar{t}$  cross section ratio in proton-proton collisions at  $\sqrt{s} = 13$  TeV  
Hayrapetyan, Aram et al.  
[JHEP 04 \(2026\) 086](#) | [arXiv:2511.01995](#) | [DOI:10.1007/JHEP04\(2026\)086](#) | 2025-11-03
72. Characterizing the initial state and dynamical evolution in XeXe and PbPb collisions using multiparticle cumulants  
Hayrapetyan, Aram et al.  
[Phys.Lett.B 876 \(2026\) 140359](#) | [arXiv:2510.26766](#) | [DOI:10.1016/j.physletb.2026.140359](#) | 2025-10-30
73. Search for single production of a vector-like T quark decaying to a top quark and a neutral scalar boson in the lepton+jets final state in proton-proton collisions at  $\sqrt{s} = 13$  TeV  
Hayrapetyan, Aram et al.  
[arXiv:2510.25874](#) | 2025-10-29
74. Measurement of the ratio of the  $B_c^+ \rightarrow J/\psi \tau^+ \nu_\tau$  and  $B_c^+ \rightarrow J/\psi \mu^+ \nu_\mu$  branching fractions using three-prong  $\tau$  lepton decays  
Chekhovsky, Vladimir et al.  
[arXiv:2510.21559](#) | 2025-10-24
75. Search for dijet resonances with data scouting in proton-proton collisions at  $\sqrt{s}=13$  TeV  
Hayrapetyan, Aram et al.  
[JHEP 04 \(2026\) 133](#) | [arXiv:2510.21641](#) | [DOI:10.1007/JHEP04\(2026\)133](#) | 2025-10-24
76. Observation of  $tWZ$  Production at the CMS Experiment  
Hayrapetyan, Aram et al.  
[Phys.Rev.Lett. 136 \(2026\) 081802](#) | [arXiv:2510.19080](#) | [DOI:10.1103/rk6w-1pcl](#) | 2025-10-21
77. Search for dark matter production in association with bottom quarks and a lepton pair in proton-proton collisions at  $\sqrt{s} = 13$  TeV  
Chekhovsky, Vladimir et al.  
[arXiv:2510.12396](#) | 2025-10-14
78. Performance of heavy-flavour jet identification in Lorentz-boosted topologies in proton-proton collisions at  $\sqrt{s} = 13$  TeV  
Hayrapetyan, Aram et al.  
[JINST 20 \(2025\) P11006](#) | [arXiv:2510.10228](#) | [DOI:10.1088/1748-0221/20/11/P11006](#) | 2025-10-11
79. Observation of Suppressed Charged-Particle Production in Ultrarelativistic Oxygen-Oxygen Collisions  
Hayrapetyan, Aram et al.  
[Phys.Rev.Lett. 136 \(2026\) 162301](#) | [arXiv:2510.09864](#) | [DOI:10.1103/89sf-9t1x](#) | 2025-10-10
80. Combination of searches for nonresonant Higgs boson pair production in proton-proton collisions at  $\sqrt{s}= 13$  TeV  
Hayrapetyan, Aram et al.  
[arXiv:2510.07527](#) | 2025-10-08
81. Observation of long-range collective flow in OO and NeNe collisions and implications for nuclear structure studies

Hayrapetyan, Aram et al.  
[arXiv:2510.02580](#) | 2025-10-02

82. Wasserstein normalized autoencoder for anomaly detection

Hayrapetyan, Aram et al.  
[arXiv:2510.02168](#) | 2025-10-02

83. Vector boson scattering and anomalous quartic couplings in final states with  $\ell\nu qq$  or  $\ell\ell qq$  plus jets using proton-proton collisions at  $\sqrt{s}=13$  TeV

Hayrapetyan, Aram et al.  
[JHEP 03 \(2026\) 022](#) | [arXiv:2510.00118](#) | [DOI:10.1007/JHEP03\(2026\)022](#) | 2025-09-30

84. Measurement of the dineutrino system kinematic variables in dileptonic top quark pair production in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Chekhovsky, Vladimir et al.  
[arXiv:2510.00160](#) | 2025-09-30

85. Simultaneous Probe of the Charm and Bottom Quark Yukawa Couplings Using  $t\bar{t}H$  Events

Hayrapetyan, Aram et al.  
[Phys.Rev.Lett. 136 \(2026\) 011801](#) | [arXiv:2509.22535](#) | [DOI:10.1103/9nwb-splk](#) | 2025-09-26

86. Search for resonances decaying to an anomalous jet and a Higgs boson in proton-proton collisions at  $\sqrt{s}=13$  TeV

Hayrapetyan, Aram et al.  
[Eur.Phys.J.C 86 \(2026\) 137](#) | [arXiv:2509.13635](#) | [DOI:10.1140/epjc/s10052-025-15176-9](#) | 2025-09-17

87. Measurements of  $t\bar{t}W$  differential cross sections and the leptonic charge asymmetry at  $\sqrt{s}=13$  TeV

Hayrapetyan, Aram et al.  
[JHEP 03 \(2026\) 083](#) | [arXiv:2509.13512](#) | [DOI:10.1007/JHEP03\(2026\)083](#) | 2025-09-16

88. Improving missing transverse momentum estimation with a deep neural network

Hayrapetyan, Aram et al.  
[Phys.Rev.D 113 \(2026\) 072010](#) | [arXiv:2509.12012](#) | [DOI:10.1103/c4z7-tqvc](#) | 2025-09-15

89. Measurement of  $D^0$  Meson Photoproduction in Ultraperipheral Heavy Ion Collisions

Chekhovsky, Vladimir et al.  
[Phys.Rev.Lett. 136 \(2026\) 122303](#) | [arXiv:2509.08626](#) | [DOI:10.1103/lckg-sdh9](#) | 2025-09-10

90. Model-independent measurement of the Higgs boson associated production with two jets and decaying to a pair of  $W$  bosons in proton-proton collisions at  $\sqrt{s}=13$  TeV

Hayrapetyan, Aram et al.  
[JHEP 03 \(2026\) 073](#) | [arXiv:2509.07958](#) | [DOI:10.1007/JHEP03\(2026\)073](#) | 2025-09-09

91. Search for a Higgs boson produced in association with a charm quark and decaying to a  $W$  boson pair in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.  
[arXiv:2508.14988](#) | 2025-08-20

92. General search for supersymmetric particles in scenarios with compressed mass spectra using proton-proton collisions at  $\sqrt{s}=13$  TeV

Chekhovsky, Vladimir et al.  
[Phys.Rev.D 112 \(2025\) 112023](#) | [arXiv:2508.13900](#) | [DOI:10.1103/b26z-zmpy](#) | 2025-08-19

93. Search for a new scalar resonance decaying to a Higgs boson and another new scalar particle in the final state with two bottom quarks and two photons in proton-proton collisions at  $\sqrt{s}=13$  TeV



Hayrapetyan, Aram et al.

[JHEP 12 \(2025\) 178](#) | [arXiv:2508.11494](#) | [DOI:10.1007/JHEP12\(2025\)178](#) | 2025-08-15

94. Search for charged lepton flavor violating Z and Z' boson decays in proton-proton collisions at  $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 112 \(2025\) 112011](#) | [arXiv:2508.07512](#) | [DOI:10.1103/8lw4-6lyk](#) | 2025-08-11

95. Search for light pseudoscalar boson pairs produced from Higgs boson decays using the  $4\tau$  and  $2\mu 2\tau$  final states in proton-proton collisions at  $\sqrt{s}=13\text{ TeV}$

Chekhovsky, Vladimir et al.

[JHEP 04 \(2026\) 132](#) | [arXiv:2508.06947](#) | [DOI:10.1007/JHEP04\(2026\)132](#) | 2025-08-09

96. Search for b-hadron decays to long-lived particles in the CMS endcap muon detectors

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 012009](#) | [arXiv:2508.06363](#) | [DOI:10.1103/7ldn-snzn](#) | 2025-08-08

97. First Exclusive Reconstruction of the  $B^{*+}$ ,  $B^{*0}$ , and  $B_s^{*0}$  Mesons and Precise Measurement of Their Masses

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 031902](#) | [arXiv:2508.05820](#) | [DOI:10.1103/njf9-4zfv](#) | 2025-08-07

98. Search for resonant production of pairs of dijet resonances through broad mediators in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[arXiv:2507.17884](#) | 2025-07-23

99. Probing the flavour structure of dimension-6 EFT operators in multilepton final states in proton-proton collisions at  $\sqrt{s}=13\text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 03 \(2026\) 140](#) | [arXiv:2507.17498](#) | [DOI:10.1007/JHEP03\(2026\)140](#) | 2025-07-23

100. Exploring small-angle emissions in charm quark jets in proton-proton collisions at  $\sqrt{s} = 5.02\text{ TeV}$

Chekhovsky, Vladimir et al.

[arXiv:2507.13469](#) | 2025-07-17

101. Evidence of medium response to hard probes using correlations of Z bosons with hadrons in heavy ion collisions

Chekhovsky, Vladimir et al.

[Phys.Lett.B 874 \(2026\) 140120](#) | [arXiv:2507.09307](#) | [DOI:10.1016/j.physletb.2025.140120](#) | 2025-07-12

102. A method for correcting the substructure of multiprong jets using the Lund jet plane

Hayrapetyan, Aram et al.

[JHEP 11 \(2025\) 038](#) | [arXiv:2507.07775](#) | [DOI:10.1007/JHEP11\(2025\)038](#) | 2025-07-10

103. Search for heavy pseudoscalar and scalar bosons decaying to a top quark pair in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 88 \(2025\) 127801](#) | [arXiv:2507.05119](#) | [DOI:10.1088/1361-6633/ae2207](#) | 2025-07-07

104. Search for the nonresonant and resonant production of a Higgs boson in association with an additional scalar boson in the  $\gamma\gamma\tau\tau$  final state in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$

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106. Search for nonresonant new physics signals in high-mass dilepton events produced in association with b-tagged jets in proton-proton collisions at  $\sqrt{s}=13$  TeV

Chekhovsky, Vladimir et al.

[JHEP 02 \(2026\) 141](#) | [arXiv:2506.13565](#) | [DOI:10.1007/JHEP02\(2026\)141](#) | 2025-06-16

107. Search for top squarks in final states with many light-flavor jets and 0, 1, or 2 charged leptons in proton-proton collisions at  $\sqrt{s}=13$  TeV

Chekhovsky, Vladimir et al.

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108. Machine learning method for enforcing variable independence in background estimation with LHC data: ABCDisCoTEC

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109. Determination of the spin and parity of all-charm tetraquarks

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[Nature 648 \(2025\) 58](#) | [arXiv:2506.07944](#) | [DOI:10.1038/s41586-025-09711-7](#) | 2025-06-09

110. Search for the Rare Decay  $D^0 \rightarrow \mu^+ \mu^-$  in Proton-Proton Collisions at  $\sqrt{s}=13.6$  TeV

Chekhovsky, Vladimir et al.

[Phys.Rev.Lett. 135 \(2025\) 151803](#) | [arXiv:2506.06152](#) | [DOI:10.1103/zc76-rgcp](#) | 2025-06-06

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Chekhovsky, Vladimir et al.

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112. Search for CP violation in events with top quarks and Z bosons at  $\sqrt{s} = 13$  and 13.6 TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 869 \(2025\) 139857](#) | [arXiv:2505.21206](#) | [DOI:10.1016/j.physletb.2025.139857](#) | 2025-05-27

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Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 135 \(2025\) 091802](#) | [arXiv:2505.20483](#) | [DOI:10.1103/6z3d-zjw4](#) | 2025-05-26

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Chekhovsky, Vladimir et al.

[Phys.Rev.D 112 \(2025\) 112006](#) | [arXiv:2505.17850](#) | [DOI:10.1103/rqvb-gfb7](#) | 2025-05-23

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[JHEP 08 \(2025\) 085](#) | [arXiv:2505.05300](#) | [DOI:10.1007/JHEP08\(2025\)085](#) | 2025-05-08

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Hayrapetyan, Aram et al.

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[JHEP 04 \(2026\) 093](#) | [arXiv:2504.13081](#) | [DOI:10.1007/JHEP04\(2026\)093](#) | 2025-04-17

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[JHEP 07 \(2025\) 118](#) | [arXiv:2504.08507](#) | [DOI:10.1007/JHEP07\(2025\)118](#) | 2025-04-11

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Hayrapetyan, Aram et al.

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[Phys.Rev.Lett. 135 \(2025\) 262301](#) | [arXiv:2504.05193](#) | [DOI:10.1103/2ssw-wwyy](#) | 2025-04-07

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[Rept.Prog.Phys. 88 \(2025\) 087801](#) | [arXiv:2503.22382](#) | [DOI:10.1088/1361-6633/adf7d3](#) | 2025-03-28

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[Phys.Rev.D 112 \(2025\) 012009](#) | [arXiv:2503.21977](#) | [DOI:10.1103/cm24-665b](#) | 2025-03-27

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131. Search for a  $cH$  signal in the associated production of at least one charm quark with a Higgs boson in the diphoton decay channel in pp collisions at  $\sqrt{s}=13$  TeV

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[Phys.Rev.Lett. 135 \(2025\) 061803](#) | [arXiv:2412.19261](#) | [DOI:10.1103/srvm-f1h3](#) | 2024-12-26

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147. Measurement of the W boson decay branching fraction ratio  $\mathcal{B}(W \rightarrow cq)/\mathcal{B}(W \rightarrow q\bar{q}')$  in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

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148. Measurement of light-by-light scattering and the Breit-Wheeler process, and search for axion-like particles in ultraperipheral PbPb collisions at  $\sqrt{s_{NN}} = 5.02$  TeV

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[Phys.Rev.D 111 \(2025\) 112004](#) | [arXiv:2412.04357](#) | [DOI:10.1103/PhysRevD.111.112004](#) | 2024-12-05

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153. Search for heavy neutral Higgs bosons A and H in the  $t\bar{t}Z$  channel in proton-proton collisions at 13 TeV

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[Phys.Lett.B 866 \(2025\) 139568](#) | [arXiv:2412.00570](#) | [DOI:10.1016/j.physletb.2025.139568](#) | 2024-11-30

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[JINST 20 \(2025\) P04012](#) | [arXiv:2411.19749](#) | [DOI:10.1088/1748-0221/20/04/P04012](#) | 2024-11-29

155. Constraints on standard model effective field theory for a Higgs boson produced in association with W or Z bosons in the  $H \rightarrow b\bar{b}$  decay channel in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Chekhovsky, Vladimir et al.

[JHEP 03 \(2025\) 114](#) | [arXiv:2411.16907](#) | [DOI:10.1007/JHEP03\(2025\)114](#) | 2024-11-25

156. Search for rare decays of the Z and Higgs bosons to a  $J/\psi$  or  $\psi(2S)$  meson and a photon in proton-proton collisions at  $\sqrt{s}=13$  TeV

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[Phys.Lett.B 865 \(2025\) 139462](#) | [arXiv:2411.15000](#) | [DOI:10.1016/j.physletb.2025.139462](#) | 2024-11-22

157. Angular analysis of the  $B^0 \rightarrow K^{*0} \mu^+ \mu^-$  decay in proton-proton collisions at  $\sqrt{s} = 13$  TeV

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[Phys.Lett.B 864 \(2025\) 139406](#) | [arXiv:2411.11820](#) | [DOI:10.1016/j.physletb.2025.139406](#) | 2024-11-18

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Hayrapetyan, Aram et al.

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159. Search for a heavy resonance decaying into a Z and a Higgs boson in events with an energetic jet and two electrons, two muons, or missing transverse momentum in proton-proton collisions at  $\sqrt{s} = 13$  TeV

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160. Measurements of inclusive and differential cross sections for top quark production in association with a Z boson in proton-proton collisions at  $\sqrt{s} = 13$  TeV

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161. Measurement of the inclusive  $t\bar{t}$  cross section in final states with at least one lepton and additional jets with  $302 \text{ pb}^{-1}$  of pp collisions at  $\sqrt{s} = 5.02$  TeV

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[JHEP 04 \(2025\) 099](#) | [arXiv:2410.21631](#) | [DOI:10.1007/JHEP04\(2025\)099](#) | 2024-10-28

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[JHEP 06 \(2025\) 006](#) | [arXiv:2410.21137](#) | [DOI:10.1007/JHEP06\(2025\)006](#) | 2024-10-28

163. Search for pair production of heavy particles decaying to a top quark and a gluon in the lepton+jets final state in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 85 \(2025\) 342](#) | [arXiv:2410.20601](#) | [DOI:10.1140/epjc/s10052-024-13729-y](#) | 2024-10-27

164. Search for the Higgs boson decays to a  $\rho^0$ ,  $\phi$ , or  $K^*_0$  meson and a photon in proton-proton collisions at  $\sqrt{s}=13$  TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 862 \(2025\) 139296](#) | [arXiv:2410.18289](#) | [DOI:10.1016/j.physletb.2025.139296](#) | 2024-10-23

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[JINST 19 \(2024\) P11021](#) | [arXiv:2410.17038](#) | [DOI:10.1088/1748-0221/19/11/P11021](#) | 2024-10-22

166. Search for high-mass resonances in a final state comprising a gluon and two hadronically decaying W bosons in proton-proton collisions at  $\sqrt{s} = 13$  TeV

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167. Search for heavy long-lived charged particles with large ionization energy loss in proton-proton collisions at  $\sqrt{s} = 13$  TeV

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[JHEP 04 \(2025\) 109](#) | [arXiv:2410.09164](#) | [DOI:10.1007/JHEP04\(2025\)109](#) | 2024-10-11

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[arXiv:2410.04578](#) | 2024-10-06

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[Phys.Rev.D 111 \(2025\) 072003](#) | [arXiv:2409.17770](#) | [DOI:10.1103/PhysRevD.111.072003](#) | 2024-09-26

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[Phys.Rev.D 111 \(2025\) 092014](#) | [arXiv:2409.13663](#) | [DOI:10.1103/PhysRevD.111.092014](#) | 2024-09-20

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[Phys.Rev.D 110 \(2024\) 112016](#) | [arXiv:2409.11067](#) | [DOI:10.1103/PhysRevD.110.112016](#) | 2024-09-17

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[Rept.Prog.Phys. 88 \(2025\) 037801](#) | [arXiv:2409.10806](#) | [DOI:10.1088/1361-6633/adaa13](#) | 2024-09-16

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Hayrapetyan, Aram et al.



[JHEP 02 \(2025\) 195](#) | [arXiv:2409.07258](#) | [DOI:10.1007/JHEP02\(2025\)195](#) | 2024-09-11

175. Measurement of inclusive and differential cross sections of single top quark production in association with a W boson in proton-proton collisions at  $\sqrt{s} = 13.6$  TeV

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[Phys.Lett.B 861 \(2025\) 139279](#) | [arXiv:2409.00838](#) | [DOI:10.1016/j.physletb.2025.139279](#) | 2024-09-01

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180. Search for bottom quark associated production of the standard model Higgs boson in final states with leptons in proton-proton collisions at  $\sqrt{s}=13$  TeV

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[Phys.Lett.B 860 \(2025\) 139173](#) | [arXiv:2408.01344](#) | [DOI:10.1016/j.physletb.2024.139173](#) | 2024-08-02

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[Phys.Rev.D 111 \(2025\) L051102](#) | [arXiv:2408.00678](#) | [DOI:10.1103/PhysRevD.111.L051102](#) | 2024-08-01

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[JHEP 12 \(2024\) 172](#) | [arXiv:2407.20425](#) | [DOI:10.1007/JHEP12\(2024\)172](#) | 2024-07-29

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[Phys.Lett.B 868 \(2025\) 139672](#) | [arXiv:2407.13615](#) | [DOI:10.1016/j.physletb.2025.139672](#) | 2024-07-18

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[JHEP 02 \(2025\) 097](#) | [arXiv:2407.10896](#) | [DOI:10.1007/JHEP02\(2025\)097](#) | 2024-07-15

189. Search for long-lived heavy neutral leptons in proton-proton collision events with a lepton-jet pair associated with a secondary vertex at  $\sqrt{s} = 13$  TeV

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[JHEP 02 \(2025\) 036](#) | [arXiv:2407.10717](#) | [DOI:10.1007/JHEP02\(2025\)036](#) | 2024-07-15

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[Phys.Rev.D 110 \(2024\) 092002](#) | [arXiv:2407.03223](#) | [DOI:10.1103/PhysRevD.110.092002](#) | 2024-07-03

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230. Search for heavy neutral leptons in final states with electrons, muons, and hadronically decaying tau leptons in proton-proton collisions at  $\sqrt{s} = 13 \text{ TeV}$

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[JHEP 06 \(2024\) 123](#) | [arXiv:2403.00100](#) | [DOI:10.1007/JHEP06\(2024\)123](#) | 2024-02-29

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[Phys.Rev.D 110 \(2024\) 012004](#) | [arXiv:2402.18658](#) | [DOI:10.1103/PhysRevD.110.012004](#) | 2024-02-28

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[Phys.Rev.Lett. 132 \(2024\) 241802](#) | [arXiv:2402.18461](#) | [DOI:10.1103/PhysRevLett.132.241802](#) | 2024-02-28

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[Comput.Softw.Big Sci. 8 \(2024\) 17](#) | [arXiv:2402.15366](#) | [DOI:10.1007/s41781-024-00124-1](#) | 2024-02-23

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[Phys.Rev.Lett. 133 \(2024\) 071903](#) | [arXiv:2402.13864](#) | [DOI:10.1103/PhysRevLett.133.071903](#) | 2024-02-21

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[Phys.Rev.D 110 \(2024\) 052004](#) | [arXiv:2402.13808](#) | [DOI:10.1103/PhysRevD.110.052004](#) | 2024-02-21

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240. Search for a scalar or pseudoscalar dilepton resonance produced in association with a massive vector boson or top quark-antiquark pair in multilepton events at  $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 110 \(2024\) 012013](#) | [arXiv:2402.11098](#) | [DOI:10.1103/PhysRevD.110.012013](#) | 2024-02-16

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[Phys.Rev.Lett. 134 \(2025\) 131802](#) | [arXiv:2402.09932](#) | [DOI:10.1103/PhysRevLett.134.131802](#) | 2024-02-15

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[Phys.Rev.Lett. 132 \(2024\) 261902](#) | [arXiv:2402.08713](#) | [DOI:10.1103/PhysRevLett.132.261902](#) | 2024-02-13

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[JHEP 02 \(2025\) 064](#) | [arXiv:2402.08486](#) | [DOI:10.1007/JHEP02\(2025\)064](#) | 2024-02-13

244. Search for pair production of scalar and vector leptoquarks decaying to muons and bottom quarks in proton-proton collisions at  $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 109 \(2024\) 112003](#) | [arXiv:2402.08668](#) | [DOI:10.1103/PhysRevD.109.112003](#) | 2024-02-09

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[Phys.Rev.D 110 \(2024\) 032007](#) | [arXiv:2402.01898](#) | [DOI:10.1103/PhysRevD.110.032007](#) | 2024-02-02

246. Combined search for electroweak production of winos, binos, higgsinos, and sleptons in proton-proton collisions at  $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 109 \(2024\) 112013](#) | [arXiv:2401.14494](#) | [DOI:10.1103/PhysRevD.109.112013](#) | 2024-01-25

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250. Test of lepton flavor universality in  $B^{\pm} \rightarrow K^{\pm} \mu^+ \mu^-$  and  $B^{\pm} \rightarrow K^{\pm} e^+ e^-$  decays in proton-proton collisions at  $\sqrt{s} = 13 \text{ TeV}$

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[Rept.Prog.Phys. 87 \(2024\) 077802](#) | [arXiv:2401.07090](#) | [DOI:10.1088/1361-6633/ad4e65](#) | 2024-01-13

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252. Observation of Enhanced Long-Range Elliptic Anisotropies Inside High-Multiplicity Jets in pp Collisions at  $\sqrt{s}=13\text{TeV}$

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[Nature Commun. 16 \(2025\) 7990](#) | [arXiv:2312.17092](#) | [DOI:10.1038/s41467-025-56200-6](#) | 2023-12-28

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[Eur.Phys.J.C 85 \(2025\) 72](#) | [arXiv:2312.16669](#) | [DOI:10.1140/epjc/s10052-024-13606-8](#) | 2023-12-27

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[JHEP 05 \(2024\) 116](#) | [arXiv:2312.16343](#) | [DOI:10.1007/JHEP05\(2024\)116](#) | 2023-12-26

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[Phys.Lett.B 855 \(2024\) 138815](#) | [arXiv:2312.11668](#) | [DOI:10.1016/j.physletb.2024.138815](#) | 2023-12-18

257. Search for Long-Lived Heavy Neutral Leptons with Lepton Flavour Conserving or Violating Decays to a Jet and a Charged Lepton

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[JHEP 03 \(2024\) 105](#) | [arXiv:2312.07484](#) | [DOI:10.1007/JHEP03\(2024\)105](#) | 2023-12-12

258. Search for flavor changing neutral current interactions of the top quark in final states with a photon and additional jets in proton-proton collisions at  $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 109 \(2024\) 072004](#) | [arXiv:2312.08229](#) | [DOI:10.1103/PhysRevD.109.072004](#) | 2023-12-11

259. Measurement of simplified template cross sections of the Higgs boson produced in association with W or Z bosons in the  $H\rightarrow b\bar{b}$  decay channel in proton-proton collisions at  $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 109 \(2024\) 092011](#) | [arXiv:2312.07562](#) | [DOI:10.1103/PhysRevD.109.092011](#) | 2023-12-09

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Hayrapetyan, Aram et al.

[Phys.Rev.D 111 \(2025\) 012009](#) | [arXiv:2312.03199](#) | [DOI:10.1103/PhysRevD.111.012009](#) | 2023-12-05

261. Search for the lepton flavor violating  $\tau \rightarrow 3\mu$  decay in proton-proton collisions at  $\sqrt{s}=13\text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Lett.B 853 \(2024\) 138633](#) | [arXiv:2312.02371](#) | [DOI:10.1016/j.physletb.2024.138633](#) | 2023-12-04

262. Higher-order moments of the elliptic flow distribution in PbPb collisions at  $\sqrt{s_{\text{NN}}} = 5.02\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 2024 \(2024\) 106](#) | [arXiv:2311.11370](#) | [DOI:10.1007/JHEP02\(2024\)106](#) | 2023-11-19

263. Search for new Higgs bosons via same-sign top quark pair production in association with a jet in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Lett.B 850 \(2024\) 138478](#) | [arXiv:2311.03261](#) | [DOI:10.1016/j.physletb.2024.138478](#) | 2023-11-06

264. Search for high-mass exclusive diphoton production with tagged protons in proton-proton collisions at  $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.D 110 \(2024\) 012010](#) | [arXiv:2311.02725](#) | [DOI:10.1103/PhysRevD.110.012010](#) | 2023-11-05

265. Search for an exotic decay of the Higgs boson into a Z boson and a pseudoscalar particle in proton-proton collisions at  $\sqrt{s}=13\text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Lett.B 852 \(2024\) 138582](#) | [arXiv:2311.00130](#) | [DOI:10.1016/j.physletb.2024.138582](#) | 2023-10-31

266. Search for  $W'$  bosons decaying to a top and a bottom quark in leptonic final states in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 046](#) | [arXiv:2310.19893](#) | [DOI:10.1007/JHEP05\(2024\)046](#) | 2023-10-30

267. Search for Narrow Trijet Resonances in Proton-Proton Collisions at  $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 133 \(2024\) 011801](#) | [arXiv:2310.14023](#) | [DOI:10.1103/PhysRevLett.133.011801](#) | 2023-10-21

268. Search for central exclusive production of top quark pairs in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$  with tagged protons

Tumasyan, Armen et al.

[JHEP 06 \(2024\) 187](#) | [arXiv:2310.11231](#) | [DOI:10.1007/JHEP06\(2024\)187](#) | 2023-10-17



269. Observation of  $WW\gamma$  production and search for  $H\gamma$  production in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 121901](#) | [arXiv:2310.05164](#) | [DOI:10.1103/PhysRevLett.132.121901](#) | 2023-10-08

270. Muon identification using multivariate techniques in the CMS experiment in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[JINST 19 \(2024\) P02031](#) | [arXiv:2310.03844](#) | [DOI:10.1088/1748-0221/19/02/P02031](#) | 2023-10-05

271. Study of azimuthal anisotropy of  $\phi(1S)$  mesons in pPb collisions at  $\sqrt{s_{NN}} = 8.16$  TeV

Tumasyan, Armen et al.

[Phys.Lett.B 850 \(2024\) 138518](#) | [arXiv:2310.03233](#) | [DOI:10.1016/j.physletb.2024.138518](#) | 2023-10-04

272. Search for stealth supersymmetry in final states with two photons, jets, and low missing transverse momentum in proton-proton collisions at  $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112009](#) | [arXiv:2310.03154](#) | [DOI:10.1103/PhysRevD.109.112009](#) | 2023-10-04

273. Search for a new resonance decaying into two spin-0 bosons in a final state with two photons and two bottom quarks in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 05 \(2024\) 316](#) | [arXiv:2310.01643](#) | [DOI:10.1007/JHEP05\(2024\)316](#) | 2023-10-02

274. Search for supersymmetry in final states with disappearing tracks in proton-proton collisions at  $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 072007](#) | [arXiv:2309.16823](#) | [DOI:10.1103/PhysRevD.109.072007](#) | 2023-09-28

275. Search for direct production of GeV-scale resonances decaying to a pair of muons in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[JHEP 12 \(2023\) 070](#) | [arXiv:2309.16003](#) | [DOI:10.1007/JHEP12\(2023\)070](#) | 2023-09-27

276. Inclusive and differential cross section measurements of  $t\bar{t}b\bar{b}$  production in the lepton+jets channel at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 042](#) | [arXiv:2309.14442](#) | [DOI:10.1007/JHEP05\(2024\)042](#) | 2023-09-25

277. Measurement of the  $\tau$  lepton polarization in Z boson decays in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[JHEP 01 \(2024\) 101](#) | [arXiv:2309.12408](#) | [DOI:10.1007/JHEP01\(2024\)101](#) | 2023-09-21

278. Development of the CMS detector for the CERN LHC Run 3

Hayrapetyan, Aram et al.

[JINST 19 \(2024\) P05064](#) | [arXiv:2309.05466](#) | [DOI:10.1088/1748-0221/19/05/P05064](#) | 2023-09-11

279. Evidence for the Higgs Boson Decay to a Z Boson and a Photon at the LHC

Aad, Georges et al.

[Phys.Rev.Lett. 132 \(2024\) 021803](#) | [arXiv:2309.03501](#) | [DOI:10.1103/PhysRevLett.132.021803](#) | 2023-09-07

280. Luminosity determination using Z boson production at the CMS experiment

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 26](#) | [arXiv:2309.01008](#) | [DOI:10.1140/epjc/s10052-023-12268-2](#) | 2023-09-02

281. Search for a third-generation leptoquark coupled to a  $\tau$  lepton and a b quark through single, pair, and nonresonant production in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 311](#) | [arXiv:2308.07826](#) | [DOI:10.1007/JHEP05\(2024\)311](#) | 2023-08-15

282. Search for Scalar Leptoquarks Produced via  $\tau$ -Lepton-Quark Scattering in pp Collisions at  $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 061801](#) | [arXiv:2308.06143](#) | [DOI:10.1103/PhysRevLett.132.061801](#) | 2023-08-11

283. Measurement of the production cross section for a W boson in association with a charm quark in proton-proton collisions at  $\sqrt{s} = 13 \text{ TeV}$

Tumasyan, Armen et al.

[Eur.Phys.J.C 84 \(2024\) 27](#) | [arXiv:2308.02285](#) | [DOI:10.1140/epjc/s10052-023-12258-4](#) | 2023-08-04

284. Measurement of the Higgs boson production via vector boson fusion and its decay into bottom quarks in proton-proton collisions at  $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 01 \(2024\) 173](#) | [arXiv:2308.01253](#) | [DOI:10.1007/JHEP01\(2024\)173](#) | 2023-08-02

285. Search for new physics in multijet events with at least one photon and large missing transverse momentum in proton-proton collisions at 13 TeV

Hayrapetyan, Aram et al.

[JHEP 10 \(2023\) 046](#) | [arXiv:2307.16216](#) | [DOI:10.1007/JHEP10\(2023\)046](#) | 2023-07-30

286. Search for physics beyond the standard model in top quark production with additional leptons in the context of effective field theory

Hayrapetyan, Aram et al.

[JHEP 12 \(2023\) 068](#) | [arXiv:2307.15761](#) | [DOI:10.1007/JHEP12\(2023\)068](#) | 2023-07-28

287. Study of charm hadronization with prompt  $\Lambda_c^+$  baryons in proton-proton and lead-lead collisions at  $\sqrt{s_{NN}} = 5.02 \text{ TeV}$

Tumasyan, Armen et al.

[JHEP 01 \(2024\) 128](#) | [arXiv:2307.11186](#) | [DOI:10.1007/JHEP01\(2024\)128](#) | 2023-07-20

288. Multiplicity and transverse momentum dependence of charge-balance functions in pPb and PbPb collisions at LHC energies

Tumasyan, Armen et al.

[JHEP 08 \(2024\) 148](#) | [arXiv:2307.11185](#) | [DOI:10.1007/JHEP08\(2024\)148](#) | 2023-07-20

289. Search for a high-mass dimuon resonance produced in association with b quark jets at  $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 10 \(2023\) 043](#) | [arXiv:2307.08708](#) | [DOI:10.1007/JHEP10\(2023\)043](#) | 2023-07-14

290. Search for  $Z'$  bosons decaying to pairs of heavy Majorana neutrinos in proton-proton collisions at  $\sqrt{s} = 13 \text{ TeV}$

Tumasyan, Armen et al.

[JHEP 11 \(2023\) 181](#) | [arXiv:2307.06959](#) | [DOI:10.1007/JHEP11\(2023\)181](#) | 2023-07-12

291. Two-particle Bose-Einstein correlations and their Lévy parameters in PbPb collisions at  $\sqrt{s_{NN}}=5.02 \text{ TeV}$

Tumasyan, Armen et al.

[Phys.Rev.C 109 \(2024\) 024914](#) | [arXiv:2306.11574](#) | [DOI:10.1103/PhysRevC.109.024914](#) | 2023-06-20

292. Performance of the local reconstruction algorithms for the CMS hadron calorimeter with Run 2 data

Tumasyan, Armen et al.

[JINST 18 \(2023\) P11017](#) | [arXiv:2306.10355](#) | [DOI:10.1088/1748-0221/18/11/P11017](#) | 2023-06-17

293. New Structures in the  $J/\psi/\psi$  Mass Spectrum in Proton-Proton Collisions at  $\sqrt{s}=13\text{TeV}$   
Hayrapetyan, Aram et al.  
[Phys.Rev.Lett. 132 \(2024\) 111901](#) | [arXiv:2306.07164](#) | [DOI:10.1103/PhysRevLett.132.111901](#) | 2023-06-12
294. Search for the lepton-flavor violating decay of the Higgs boson and additional Higgs bosons in the  $e\mu$  final state in proton-proton collisions at  $\sqrt{s} = 13 \text{ TeV}$   
Hayrapetyan, Aram et al.  
[Phys.Rev.D 108 \(2023\) 072004](#) | [arXiv:2305.18106](#) | [DOI:10.1103/PhysRevD.108.072004](#) | 2023-05-29
295. Measurements of the azimuthal anisotropy of prompt and nonprompt charmonia in PbPb collisions at  $\sqrt{s_{\text{NN}}} = 5.02 \text{ TeV}$   
Tumasyan, Armen et al.  
[JHEP 10 \(2023\) 115](#) | [arXiv:2305.16928](#) | [DOI:10.1007/JHEP10\(2023\)115](#) | 2023-05-26
296. Observation of four top quark production in proton-proton collisions at  $\sqrt{s}=13 \text{ TeV}$   
Hayrapetyan, Aram et al.  
[Phys.Lett.B 847 \(2023\) 138290](#) | [arXiv:2305.13439](#) | [DOI:10.1016/j.physletb.2023.138290](#) | 2023-05-22
297. Search for Inelastic Dark Matter in Events with Two Displaced Muons and Missing Transverse Momentum in Proton-Proton Collisions at  $\sqrt{s}=13\text{TeV}$   
Hayrapetyan, Aram et al.  
[Phys.Rev.Lett. 132 \(2024\) 041802](#) | [arXiv:2305.11649](#) | [DOI:10.1103/PhysRevLett.132.041802](#) | 2023-05-19
298. Search for resonances in events with photon and jet final states in proton-proton collisions at  $\sqrt{s} = 13 \text{ TeV}$   
Tumasyan, Armen et al.  
[JHEP 12 \(2023\) 189](#) | [arXiv:2305.07998](#) | [DOI:10.1007/JHEP12\(2023\)189](#) | 2023-05-13
299. Measurements of inclusive and differential cross sections for the Higgs boson production and decay to four-leptons in proton-proton collisions at  $\sqrt{s} = 13 \text{ TeV}$   
Hayrapetyan, Aram et al.  
[JHEP 08 \(2023\) 040](#) | [arXiv:2305.07532](#) | [DOI:10.1007/JHEP08\(2023\)040](#) | 2023-05-12
300. Observation of the rare decay of the  $\eta$  meson to four muons  
Hayrapetyan, Aram et al.  
[Phys.Rev.Lett. 131 \(2023\) 091903](#) | [arXiv:2305.04904](#) | [DOI:10.1103/PhysRevLett.131.091903](#) | 2023-05-08
301. Search for top squark pair production in a final state with at least one hadronically decaying tau lepton in proton-proton collisions at  $\sqrt{s} = 13 \text{ TeV}$   
Tumasyan, Armen et al.  
[JHEP 07 \(2023\) 110](#) | [arXiv:2304.07174](#) | [DOI:10.1007/JHEP07\(2023\)110](#) | 2023-04-14
302. Probing Small Bjorken- $x$  Nuclear Gluonic Structure via Coherent  $J/\psi$  Photoproduction in Ultraperipheral Pb-Pb Collisions at  $\sqrt{s_{\text{NN}}}=5.02\text{TeV}$   
Tumasyan, Armen et al.  
[Phys.Rev.Lett. 131 \(2023\) 262301](#) | [arXiv:2303.16984](#) | [DOI:10.1103/PhysRevLett.131.262301](#) | 2023-03-29
303. Observation of the  $\chi(3S)$  Meson and Suppression of  $\chi$  States in Pb-Pb Collisions at  $\sqrt{s_{\text{NN}}}=5.02\text{TeV}$   
Tumasyan, Armen et al.  
[Phys.Rev.Lett. 133 \(2024\) 022302](#) | [arXiv:2303.17026](#) | [DOI:10.1103/PhysRevLett.133.022302](#) | 2023-03-29
304. First measurement of the top quark pair production cross section in proton-proton collisions at  $\sqrt{s} = 13.6 \text{ TeV}$   
Tumasyan, Armen et al.  
[JHEP 08 \(2023\) 204](#) | [arXiv:2303.10680](#) | [DOI:10.1007/JHEP08\(2023\)204](#) | 2023-03-19

305. A search for new physics in central exclusive production using the missing mass technique with the CMS detector and the CMS-TOTEM precision proton spectrometer  
Tumasyan, Armen et al.  
[Eur.Phys.J.C 83 \(2023\) 827](#) | [arXiv:2303.04596](#) | [DOI:10.1140/epjc/s10052-023-11687-5](#) | 2023-03-08
306. Evidence for Four-Top Quark Production in Proton-Proton Collisions at  $\sqrt{s}=13$  TeV  
Tumasyan, Armen et al.  
[Phys.Lett.B 844 \(2023\) 138076](#) | [arXiv:2303.03864](#) | [DOI:10.1016/j.physletb.2023.138076](#) | 2023-03-07
307. A search for decays of the Higgs boson to invisible particles in events with a top-antitop quark pair or a vector boson in proton-proton collisions at  $\sqrt{s} = 13$  TeV  
Tumasyan, Armen et al.  
[Eur.Phys.J.C 83 \(2023\) 933](#) | [arXiv:2303.01214](#) | [DOI:10.1140/epjc/s10052-023-11952-7](#) | 2023-03-02
308. Search for dark matter particles in  $W^+W^-$  events with transverse momentum imbalance in proton-proton collisions at  $\sqrt{s} = 13$  TeV  
Hayrapetyan, Aram et al.  
[JHEP 03 \(2024\) 134](#) | [arXiv:2310.12229](#) | [DOI:10.1007/JHEP03\(2024\)134](#) | 2023-03
309. Search for a vector-like quark  $T' \rightarrow tH$  via the diphoton decay mode of the Higgs boson in proton-proton collisions at  $\sqrt{s} = 13$  TeV  
Tumasyan, Armen et al.  
[JHEP 09 \(2023\) 057](#) | [arXiv:2302.12802](#) | [DOI:10.1007/JHEP09\(2023\)057](#) | 2023-02-24
310. Measurement of the top quark mass using a profile likelihood approach with the lepton + jets final states in proton-proton collisions at  $\sqrt{s}=13$  TeV  
Tumasyan, Armen et al.  
[Eur.Phys.J.C 83 \(2023\) 963](#) | [arXiv:2302.01967](#) | [DOI:10.1140/epjc/s10052-023-12050-4](#) | 2023-02-03
311. Search for top squarks in the four-body decay mode with single lepton final states in proton-proton collisions at  $\sqrt{s} = 13$  TeV  
Tumasyan, Armen et al.  
[JHEP 06 \(2023\) 060](#) | [arXiv:2301.08096](#) | [DOI:10.1007/JHEP06\(2023\)060](#) | 2023-01-19
312. First Measurement of the Forward Rapidity Gap Distribution in pPb Collisions at  $\sqrt{s_{NN}}=8.16$  TeV  
Tumasyan, Armen et al.  
[Phys.Rev.D 108 \(2023\) 092004](#) | [arXiv:2301.07630](#) | [DOI:10.1103/PhysRevD.108.092004](#) | 2023-01-18
313.  $K_S^0$  and  $\Lambda(\Lambda^0)$  two-particle femtoscopic correlations in PbPb collisions at  $\sqrt{s_{NN}}=5.02$  TeV  
Tumasyan, Armen et al.  
[Phys.Lett.B 857 \(2024\) 138936](#) | [arXiv:2301.05290](#) | [DOI:10.1016/j.physletb.2024.138936](#) | 2023-01-12
314. Search for new physics in the  $\tau$  lepton plus missing transverse momentum final state in proton-proton collisions at  $\sqrt{s} = 13$  TeV  
Tumasyan, A. et al.  
[JHEP 09 \(2023\) 051](#) | [arXiv:2212.12604](#) | [DOI:10.1007/JHEP09\(2023\)051](#) | 2022-12-23
315. Measurement of the electroweak production of  $W\gamma$  in association with two jets in proton-proton collisions at  $\sqrt{s} = 13$  TeV  
Tumasyan, Armen et al.  
[Phys.Rev.D 108 \(2023\) 032017](#) | [arXiv:2212.12592](#) | [DOI:10.1103/PhysRevD.108.032017](#) | 2022-12-23
316. Measurement of the  $B^0_S \rightarrow \mu^+ \mu^-$  decay properties and search for the  $B^0 \rightarrow \mu^+ \mu^-$  decay in proton-proton collisions at  $\sqrt{s} = 13$  TeV  
Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137955](#) | [arXiv:2212.10311](#) | [DOI:10.1016/j.physletb.2023.137955](#) | 2022-12-20

317. Search for long-lived particles using out-of-time trackless jets in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 210](#) | [arXiv:2212.06695](#) | [DOI:10.1007/JHEP07\(2023\)210](#) | 2022-12-13

318. Measurement of the dependence of the hadron production fraction ratio  $f_s / f_u$  and  $f_d / f_u$  on B meson kinematic variables in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 121901](#) | [arXiv:2212.02309](#) | [DOI:10.1103/PhysRevLett.131.121901](#) | 2022-12-05

319. Measurements of azimuthal anisotropy of nonprompt D0 mesons in PbPb collisions at  $\sqrt{s_{NN}}=5.02$  TeV

Tumasyan, Armen et al.

[Phys.Lett.B 850 \(2024\) 138389](#) | [arXiv:2212.01636](#) | [DOI:10.1016/j.physletb.2023.138389](#) | 2022-12-03

320. Search for high-mass exclusive  $\gamma\gamma \rightarrow WW$  and  $\gamma\gamma \rightarrow ZZ$  production in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 229](#) | [arXiv:2211.16320](#) | [DOI:10.1007/JHEP07\(2023\)229](#) | 2022-11-29

321. Search for Higgs Boson and Observation of Z Boson through their Decay into a Charm Quark-Antiquark Pair in Boosted Topologies in Proton-Proton Collisions at  $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 041801](#) | [arXiv:2211.14181](#) | [DOI:10.1103/PhysRevLett.131.041801](#) | 2022-11-25

322. Search for supersymmetry in final states with a single electron or muon using angular correlations and heavy-object identification in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 09 \(2023\) 149](#) | [arXiv:2211.08476](#) | [DOI:10.1007/JHEP09\(2023\)149](#) | 2022-11-15

323. Measurement of the differential tt production cross section as a function of the jet mass and extraction of the top quark mass in hadronic decays of boosted top quarks

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 560](#) | [arXiv:2211.01456](#) | [DOI:10.1140/epjc/s10052-023-11587-8](#) | 2022-11-02

324. Azimuthal correlations in Z +jets events in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 722](#) | [arXiv:2210.16139](#) | [DOI:10.1140/epjc/s10052-023-11833-z](#) | 2022-10-28

325. Measurements of jet multiplicity and jet transverse momentum in multijet events in proton-proton collisions at  $\sqrt{s}=13$  TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 742](#) | [arXiv:2210.13557](#) | [DOI:10.1140/epjc/s10052-023-11753-y](#) | 2022-10-24

326. Search for medium effects using jets from bottom quarks in PbPb collisions at  $\sqrt{s_{NN}}=5.02$  TeV

Tumasyan, Armen et al.

[Phys.Lett.B 844 \(2023\) 137849](#) | [arXiv:2210.08547](#) | [DOI:10.1016/j.physletb.2023.137849](#) | 2022-10-16

327. Azimuthal anisotropy of dijet events in PbPb collisions at  $\sqrt{s_{NN}} = 5.02$  TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 139](#) | [arXiv:2210.08325](#) | [DOI:10.1007/JHEP07\(2023\)139](#) | 2022-10-15

328. Proton reconstruction with the CMS-TOTEM Precision Proton Spectrometer

Tumasyan, Armen et al.

[JINST 18 \(2023\) P09009](#) | [arXiv:2210.05854](#) | [DOI:10.1088/1748-0221/18/09/P09009](#) | 2022-10-11

329. Search for a heavy composite Majorana neutrino in events with dilepton signatures from proton-proton collisions at  $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 843 \(2023\) 137803](#) | [arXiv:2210.03082](#) | [DOI:10.1016/j.physletb.2023.137803](#) | 2022-10-06

330. Search for new heavy resonances decaying to WW, WZ, ZZ, WH, or ZH boson pairs in the all-jets final state in proton-proton collisions at  $\sqrt{s}=13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 844 \(2023\) 137813](#) | [arXiv:2210.00043](#) | [DOI:10.1016/j.physletb.2023.137813](#) | 2022-09-30

331. Search for pair production of vector-like quarks in leptonic final states in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 020](#) | [arXiv:2209.07327](#) | [DOI:10.1007/JHEP07\(2023\)020](#) | 2022-09-15

332. Search for exotic Higgs boson decays  $H \rightarrow \mathcal{A}\mathcal{A} \rightarrow 4\gamma$  with events containing two merged diphotons in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 101801](#) | [arXiv:2209.06197](#) | [DOI:10.1103/PhysRevLett.131.101801](#) | 2022-09-13

333. Search for new physics using effective field theory in 13 TeV pp collision events that contain a top quark pair and a boosted Z or Higgs boson

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 032008](#) | [arXiv:2208.12837](#) | [DOI:10.1103/PhysRevD.108.032008](#) | 2022-08-26

334. Measurement of the Higgs boson inclusive and differential fiducial production cross sections in the diphoton decay channel with pp collisions at  $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 091](#) | [arXiv:2208.12279](#) | [DOI:10.1007/JHEP07\(2023\)091](#) | 2022-08-25

335. Search for pair-produced vector-like leptons in final states with third-generation leptons and at least three b quark jets in proton-proton collisions at  $\sqrt{s}=13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 846 \(2023\) 137713](#) | [arXiv:2208.09700](#) | [DOI:10.1016/j.physletb.2023.137713](#) | 2022-08-20

336. Measurement of the cross section of top quark-antiquark pair production in association with a W boson in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 219](#) | [arXiv:2208.06485](#) | [DOI:10.1007/JHEP07\(2023\)219](#) | 2022-08-12

337. Search for CP violation in ttH and tH production in multilepton channels in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 092](#) | [arXiv:2208.02686](#) | [DOI:10.1007/JHEP07\(2023\)092](#) | 2022-08-04

338. Measurement of the  $t\bar{t}$  charge asymmetry in events with highly Lorentz-boosted top quarks in pp collisions at  $\sqrt{s}=13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 846 \(2023\) 137703](#) | [arXiv:2208.02751](#) | [DOI:10.1016/j.physletb.2023.137703](#) | 2022-08-04

339. Searches for additional Higgs bosons and for vector leptoquarks in  $\tau\tau$  final states in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 073](#) | [arXiv:2208.02717](#) | [DOI:10.1007/JHEP07\(2023\)073](#) | 2022-08-04



340. Search for the exotic decay of the Higgs boson into two light pseudoscalars with four photons in the final state in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 148](#) | [arXiv:2208.01469](#) | [DOI:10.1007/JHEP07\(2023\)148](#) | 2022-08-02

341. Measurement of inclusive and differential cross sections for single top quark production in association with a W boson in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 046](#) | [arXiv:2208.00924](#) | [DOI:10.1007/JHEP07\(2023\)046](#) | 2022-08-01

342. Search for the Higgs boson decay to a pair of electrons in proton-proton collisions at  $\sqrt{s}=13$  TeV

Tumasyan, Armen et al.

[Phys.Lett.B 846 \(2023\) 137783](#) | [arXiv:2208.00265](#) | [DOI:10.1016/j.physletb.2023.137783](#) | 2022-07-30

343. Search for new physics in the lepton plus missing transverse momentum final state in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 07 \(2022\) 067](#) | [arXiv:2202.06075](#) | [DOI:10.1007/JHEP07\(2022\)067](#) | 2022-07-11

344. Measurement of the top quark pole mass using  $t\bar{t}$ +jet events in the dilepton final state in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 077](#) | [arXiv:2207.02270](#) | [DOI:10.1007/JHEP07\(2023\)077](#) | 2022-07-05

345. Search for direct pair production of supersymmetric partners of  $\tau$  leptons in the final state with two hadronically decaying  $\tau$  leptons and missing transverse momentum in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 012011](#) | [arXiv:2207.02254](#) | [DOI:10.1103/PhysRevD.108.012011](#) | 2022-07-05

346. Search for a charged Higgs boson decaying into a heavy neutral Higgs boson and a W boson in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 09 \(2023\) 032](#) | [arXiv:2207.01046](#) | [DOI:10.1007/JHEP09\(2023\)032](#) | 2022-07-03

347. Search for Higgs boson pairs decaying to  $WW^*WW^*$ ,  $WW^*\tau\tau$ , and  $\tau\tau\tau\tau$  in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 095](#) | [arXiv:2206.10268](#) | [DOI:10.1007/JHEP07\(2023\)095](#) | 2022-06-21

348. Search for nonresonant Higgs boson pair production in the four leptons plus two b jets final state in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 06 \(2023\) 130](#) | [arXiv:2206.10657](#) | [DOI:10.1007/JHEP06\(2023\)130](#) | 2022-06-21

349. Search for nonresonant Higgs boson pair production in final state with two bottom quarks and two tau leptons in proton-proton collisions at  $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137531](#) | [arXiv:2206.09401](#) | [DOI:10.1016/j.physletb.2022.137531](#) | 2022-06-19

350. Measurements of the Higgs boson production cross section and couplings in the W boson pair decay channel in proton-proton collisions at  $\sqrt{s}=13$  TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 667](#) | [arXiv:2206.09466](#) | [DOI:10.1140/epjc/s10052-023-11632-6](#) | 2022-06-19



351. Probing Heavy Majorana Neutrinos and the Weinberg Operator through Vector Boson Fusion Processes in Proton-Proton Collisions at  $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 011803](#) | [arXiv:2206.08956](#) | [DOI:10.1103/PhysRevLett.131.011803](#) | 2022-06-17

352. Precision measurement of the Z boson invisible width in pp collisions at  $\sqrt{s}=13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137563](#) | [arXiv:2206.07110](#) | [DOI:10.1016/j.physletb.2022.137563](#) | 2022-06-14

353. Observation of  $\tau$  lepton pair production in ultraperipheral lead-lead collisions at  $\sqrt{s_{\text{NN}}} = 5.02\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 151803](#) | [arXiv:2206.05192](#) | [DOI:10.1103/PhysRevLett.131.151803](#) | 2022-06-10

354. Search for Higgs boson decays into Z and  $J/\psi$  and for Higgs and Z boson decays into  $J/\psi$  or Y pairs in pp collisions at  $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137534](#) | [arXiv:2206.03525](#) | [DOI:10.1016/j.physletb.2022.137534](#) | 2022-06-07

355. Observation of same-sign WW production from double parton scattering in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 091803](#) | [arXiv:2206.02681](#) | [DOI:10.1103/PhysRevLett.131.091803](#) | 2022-06-06

356. Combination of inclusive top-quark pair production cross-section measurements using ATLAS and CMS data at  $\sqrt{s} = 7$  and  $8\text{ TeV}$

Aad, G. et al.

[JHEP 07 \(2023\) 213](#) | [arXiv:2205.13830](#) | [DOI:10.1007/JHEP07\(2023\)213](#) | 2022-05-27

357. Search for electroweak production of charginos and neutralinos at  $\sqrt{s}=13\text{ TeV}$  in final states containing hadronic decays of WW, WZ, or WH and missing transverse momentum

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137460](#) | [arXiv:2205.09597](#) | [DOI:10.1016/j.physletb.2022.137460](#) | 2022-05-19

358. Search for long-lived particles decaying to a pair of muons in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 05 \(2023\) 228](#) | [arXiv:2205.08582](#) | [DOI:10.1007/JHEP05\(2023\)228](#) | 2022-05-17

359. Search for CP violating top quark couplings in pp collisions at  $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 023](#) | [arXiv:2205.07434](#) | [DOI:10.1007/JHEP07\(2023\)023](#) | 2022-05-15

360. Search for heavy resonances and quantum black holes in  $e\mu$ ,  $e\tau$ , and  $\mu\tau$  final states in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 05 \(2023\) 227](#) | [arXiv:2205.06709](#) | [DOI:10.1007/JHEP05\(2023\)227](#) | 2022-05-13

361. Search for Nonresonant Pair Production of Highly Energetic Higgs Bosons Decaying to Bottom Quarks

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 041803](#) | [arXiv:2205.06667](#) | [DOI:10.1103/PhysRevLett.131.041803](#) | 2022-05-13

362. Search for Higgs Boson Decay to a Charm Quark-Antiquark Pair in Proton-Proton Collisions at  $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 061801](#) | [arXiv:2205.05550](#) | [DOI:10.1103/PhysRevLett.131.061801](#) | 2022-05-11

363. Observation of electroweak  $W+W^-$  pair production in association with two jets in proton-proton collisions at  $\sqrt{s}=13$  TeV

Tumasyan, Armen et al.

[Phys.Lett.B 841 \(2023\) 137495](#) | [arXiv:2205.05711](#) | [DOI:10.1016/j.physletb.2022.137495](#) | 2022-05-11

364. Measurement of the mass dependence of the transverse momentum of lepton pairs in Drell-Yan production in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 628](#) | [arXiv:2205.04897](#) | [DOI:10.1140/epjc/s10052-023-11631-7](#) | 2022-05-10

365. Constraints on anomalous Higgs boson couplings to vector bosons and fermions from the production of Higgs bosons using the  $\tau\tau$  final state

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 032013](#) | [arXiv:2205.05120](#) | [DOI:10.1103/PhysRevD.108.032013](#) | 2022-05-10

366. CMS pythia 8 colour reconnection tunes based on underlying-event data

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 587](#) | [arXiv:2205.02905](#) | [DOI:10.1140/epjc/s10052-023-11630-8](#) | 2022-05-05

367. Measurement of differential cross sections for the production of a Z boson in association with jets in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 052004](#) | [arXiv:2205.02872](#) | [DOI:10.1103/PhysRevD.108.052004](#) | 2022-05-05

368. Search for CP violation using tt events in the lepton+jets channel in pp collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 06 \(2023\) 081](#) | [arXiv:2205.02314](#) | [DOI:10.1007/JHEP06\(2023\)081](#) | 2022-05-04

369. Search for narrow resonances in the b-tagged dijet mass spectrum in proton-proton collisions at  $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 012009](#) | [arXiv:2205.01835](#) | [DOI:10.1103/PhysRevD.108.012009](#) | 2022-05-03

370. Strange hadron collectivity in pPb and PbPb collisions

Tumasyan, Armen et al.

[JHEP 05 \(2023\) 007](#) | [arXiv:2205.00080](#) | [DOI:10.1007/JHEP05\(2023\)007](#) | 2022-04-29

371. Azimuthal Correlations within Exclusive Dijets with Large Momentum Transfer in Photon-Lead Collisions

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 051901](#) | [arXiv:2205.00045](#) | [DOI:10.1103/PhysRevLett.131.051901](#) | 2022-04-29

372. Two-particle azimuthal correlations in  $\gamma p$  interactions using pPb collisions at  $\sqrt{s_{NN}}=8.16$  TeV

Tumasyan, Armen et al.

[Phys.Lett.B 844 \(2023\) 137905](#) | [arXiv:2204.13486](#) | [DOI:10.1016/j.physletb.2023.137905](#) | 2022-04-28

373. Search for light Higgs bosons from supersymmetric cascade decays in pp collisions at  $\sqrt{s}=13$  TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 571](#) | [arXiv:2204.13532](#) | [DOI:10.1140/epjc/s10052-023-11581-0](#) | 2022-04-28

374. Search for Higgs boson decays to a Z boson and a photon in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 05 \(2023\) 233](#) | [arXiv:2204.12945](#) | [DOI:10.1007/JHEP05\(2023\)233](#) | 2022-04-27

375. Measurements of Higgs boson production in the decay channel with a pair of  $\tau$  leptons in proton-proton collisions at  $\sqrt{s}=13$  TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 562](#) | [arXiv:2204.12957](#) | [DOI:10.1140/epjc/s10052-023-11452-8](#) | 2022-04-27

376. Search for a massive scalar resonance decaying to a light scalar and a Higgs boson in the four b quarks final state with boosted topology

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137392](#) | [arXiv:2204.12413](#) | [DOI:10.1016/j.physletb.2022.137392](#) | 2022-04-26

377. Reconstruction of decays to merged photons using end-to-end deep learning with domain continuation in the CMS detector

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 052002](#) | [arXiv:2204.12313](#) | [DOI:10.1103/PhysRevD.108.052002](#) | 2022-04-26

378. Search for new particles in an extended Higgs sector with four b quarks in the final state at  $\sqrt{s}=13$  TeV

Tumasyan, Armen et al.

[Phys.Lett.B 835 \(2022\) 137566](#) | [arXiv:2203.00480](#) | [DOI:10.1016/j.physletb.2022.137566](#) | 2022-03-01

379. Search for a  $W'$  boson decaying to a vector-like quark and a top or bottom quark in the all-jets final state at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 09 \(2022\) 088](#) | [arXiv:2202.12988](#) | [DOI:10.1007/JHEP09\(2022\)088](#) | 2022-02-25

380. Measurement of the Drell-Yan forward-backward asymmetry at high dilepton masses in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 2022 \(2022\) 063](#) | [arXiv:2202.12327](#) | [DOI:10.1007/JHEP08\(2022\)063](#) | 2022-02-24

381. Nuclear modification of  $\Upsilon$  states in pPb collisions at  $\sqrt{s_{NN}} = 5.02$  TeV

Tumasyan, Armen et al.

[Phys.Lett.B 835 \(2022\) 137397](#) | [arXiv:2202.11807](#) | [DOI:10.1016/j.physletb.2022.137397](#) | 2022-02-23

382. Search for Higgs Boson Pair Production in the Four b Quark Final State in Proton-Proton Collisions at  $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 129 \(2022\) 081802](#) | [arXiv:2202.09617](#) | [DOI:10.1103/PhysRevLett.129.081802](#) | 2022-02-19

383. Inclusive nonresonant multilepton probes of new phenomena at  $\sqrt{s}=13$  TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 112007](#) | [arXiv:2202.08676](#) | [DOI:10.1103/PhysRevD.105.112007](#) | 2022-02-17

384. Measurement of the Higgs boson width and evidence of its off-shell contributions to ZZ production

Tumasyan, Armen et al.

[Nature Phys. 18 \(2022\) 1329](#) | [arXiv:2202.06923](#) | [DOI:10.1038/s41567-022-01682-0](#) | 2022-02-14

385. Search for invisible decays of the Higgs boson produced via vector boson fusion in proton-proton collisions at  $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 092007](#) | [arXiv:2201.11585](#) | [DOI:10.1103/PhysRevD.105.092007](#) | 2022-01-27

386. Observation of  $B^0 \rightarrow \psi(2S) K^0_S \pi^+ \pi^-$  and  $B^0_S \rightarrow \psi(2S) K^0_S$  decays

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 499](#) | [arXiv:2201.09131](#) | [DOI:10.1140/epjc/s10052-022-10315-y](#) | 2022-01-22

387. Search for resonances decaying to three W bosons in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 129 \(2022\) 021802](#) | [arXiv:2201.08476](#) | [DOI:10.1103/PhysRevLett.129.021802](#) | 2022-01-20

388. Identification of hadronic tau lepton decays using a deep neural network

Tumasyan, Armen et al.

[JINST 17 \(2022\) P07023](#) | [arXiv:2201.08458](#) | [DOI:10.1088/1748-0221/17/07/P07023](#) | 2022-01-20

389. Precision measurement of the W boson decay branching fractions in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 072008](#) | [arXiv:2201.07861](#) | [DOI:10.1103/PhysRevD.105.072008](#) | 2022-01-19

390. Search for charged-lepton flavor violation in top quark production and decay in pp collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 06 \(2022\) 082](#) | [arXiv:2201.07859](#) | [DOI:10.1007/JHEP06\(2022\)082](#) | 2022-01-19

391. Measurement of the inclusive and differential  $t\bar{t}\gamma$  cross sections in the dilepton channel and effective field theory interpretation in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 091](#) | [arXiv:2201.07301](#) | [DOI:10.1007/JHEP05\(2022\)091](#) | 2022-01-18

392. Search for long-lived heavy neutral leptons with displaced vertices in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 07 \(2022\) 081](#) | [arXiv:2201.05578](#) | [DOI:10.1007/JHEP07\(2022\)081](#) | 2022-01-14

393. Search for higgsinos decaying to two Higgs bosons and missing transverse momentum in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 014](#) | [arXiv:2201.04206](#) | [DOI:10.1007/JHEP05\(2022\)014](#) | 2022-01-11

394. Observation of the  $B_c^+$  Meson in Pb-Pb and pp Collisions at  $\sqrt{s_{NN}} = 5.02$  TeV and Measurement of its Nuclear Modification Factor

Tumasyan, Armen et al.

[Phys.Rev.Lett. 128 \(2022\) 252301](#) | [arXiv:2201.02659](#) | [DOI:10.1103/PhysRevLett.128.252301](#) | 2022-01-07

395. Search for high-mass resonances decaying to a jet and a Lorentz-boosted resonance in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[Phys.Lett.B 832 \(2022\) 137263](#) | [arXiv:2201.02140](#) | [DOI:10.1016/j.physletb.2022.137263](#) | 2022-01-06

396. Search for single production of a vector-like T quark decaying to a top quark and a Z boson in the final state with jets and missing transverse momentum at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 093](#) | [arXiv:2201.02227](#) | [DOI:10.1007/JHEP05\(2022\)093](#) | 2022-01-06

397. Search for long-lived particles decaying into muon pairs in proton-proton collisions at  $\sqrt{s} = 13$  TeV collected with a dedicated high-rate data stream

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 062](#) | [arXiv:2112.13769](#) | [DOI:10.1007/JHEP04\(2022\)062](#) | 2021-12-27

398. Search for resonances decaying to three W bosons in the hadronic final state in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[Phys.Rev.D 106 \(2022\) 012002](#) | [arXiv:2112.13090](#) | [DOI:10.1103/PhysRevD.106.012002](#) | 2021-12-24

399. Probing Charm Quark Dynamics via Multiparticle Correlations in Pb-Pb Collisions at  $\sqrt{s_{NN}} = 5.02$  TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 129 \(2022\) 022001](#) | [arXiv:2112.12236](#) | [DOI:10.1103/PhysRevLett.129.022001](#) | 2021-12-22

400. Search for resonant production of strongly coupled dark matter in proton-proton collisions at 13 TeV

Tumasyan, Armen et al.

[JHEP 06 \(2022\) 156](#) | [arXiv:2112.11125](#) | [DOI:10.1007/JHEP06\(2022\)156](#) | 2021-12-21

401. Search for flavor-changing neutral current interactions of the top quark and the Higgs boson decaying to a bottom quark-antiquark pair at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 02 \(2022\) 169](#) | [arXiv:2112.09734](#) | [DOI:10.1007/JHEP02\(2022\)169](#) | 2021-12-17

402. Measurement of the production cross section for Z+b jets in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 092014](#) | [arXiv:2112.09659](#) | [DOI:10.1103/PhysRevD.105.092014](#) | 2021-12-17

403. Measurement of the inclusive tt production cross section in proton-proton collisions at  $\sqrt{s} = 5.02$  TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 144](#) | [arXiv:2112.09114](#) | [DOI:10.1007/JHEP04\(2022\)144](#) | 2021-12-16

404. Evidence for WW/WZ vector boson scattering in the decay channel  $\ell\nu qq$  produced in association with two jets in proton-proton collisions at  $\sqrt{s}=13$  TeV

Tumasyan, Armen et al.

[Phys.Lett.B 834 \(2022\) 137438](#) | [arXiv:2112.05259](#) | [DOI:10.1016/j.physletb.2022.137438](#) | 2021-12-09

405. Search for a right-handed W boson and a heavy neutrino in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 047](#) | [arXiv:2112.03949](#) | [DOI:10.1007/JHEP04\(2022\)047](#) | 2021-12-07

406. Search for heavy resonances decaying to a pair of Lorentz-boosted Higgs bosons in final states with leptons and a bottom quark pair at  $\sqrt{s}= 13$  TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 005](#) | [arXiv:2112.03161](#) | [DOI:10.1007/JHEP05\(2022\)005](#) | 2021-12-06

407. Measurements of the associated production of a W boson and a charm quark in proton-proton collisions at  $\sqrt{s}=8$  TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 1094](#) | [arXiv:2112.00895](#) | [DOI:10.1140/epjc/s10052-022-10897-7](#) | 2021-12-01

408. Measurement of  $W^{\pm}\gamma$  differential cross sections in proton-proton collisions at  $\sqrt{s} = 13$  TeV and effective field theory constraints

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 052003](#) | [arXiv:2111.13948](#) | [DOI:10.1103/PhysRevD.105.052003](#) | 2021-11-27

409. Search for heavy resonances decaying to ZZ or ZW and axion-like particles mediating nonresonant ZZ or ZH production at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 087](#) | [arXiv:2111.13669](#) | [DOI:10.1007/JHEP04\(2022\)087](#) | 2021-11-26

410. Search for a heavy resonance decaying into a top quark and a W boson in the lepton+jets final state at  $\sqrt{s} = 13$  TeV

Tumasyan, A. et al.

[JHEP 04 \(2022\) 048](#) | [arXiv:2111.10216](#) | [DOI:10.1007/JHEP04\(2022\)048](#) | 2021-11-19

411. Measurement and QCD analysis of double-differential inclusive jet cross sections in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 02 \(2022\) 142](#) | [arXiv:2111.10431](#) | [DOI:10.1007/JHEP02\(2022\)142](#) | 2021-11-19

412. Strategies and performance of the CMS silicon tracker alignment during LHC Run 2

Tumasyan, Armen et al.

[Nucl.Instrum.Meth.A 1037 \(2022\) 166795](#) | [arXiv:2111.08757](#) | [DOI:10.1016/j.nima.2022.166795](#) | 2021-11-16

413. Search for supersymmetry in final states with two or three soft leptons and missing transverse momentum in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 091](#) | [arXiv:2111.06296](#) | [DOI:10.1007/JHEP04\(2022\)091](#) | 2021-11-11

414. Observation of triple  $J/\psi$  meson production in proton-proton collisions

Tumasyan, Armen et al.

[Nature Phys. 19 \(2023\) 338](#) | [arXiv:2111.05370](#) | [DOI:10.1038/s41567-022-01838-y](#) | 2021-11-09

415. Study of dijet events with large rapidity separation in proton-proton collisions at  $\sqrt{s} = 2.76$  TeV

Tumasyan, Armen et al.

[JHEP 03 \(2022\) 189](#) | [arXiv:2111.04605](#) | [DOI:10.1007/JHEP03\(2022\)189](#) | 2021-11-08

416. A new calibration method for charm jet identification validated with proton-proton collision events at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JINST 17 \(2022\) P03014](#) | [arXiv:2111.03027](#) | [DOI:10.1088/1748-0221/17/03/P03014](#) | 2021-11-04

417. Inclusive and differential cross section measurements of single top quark production in association with a Z boson in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 02 \(2022\) 107](#) | [arXiv:2111.02860](#) | [DOI:10.1007/JHEP02\(2022\)107](#) | 2021-11-04

418. Search for Flavor-Changing Neutral Current Interactions of the Top Quark and Higgs Boson in Final States with Two Photons in Proton-Proton Collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, A. et al.

[Phys.Rev.Lett. 129 \(2022\) 032001](#) | [arXiv:2111.02219](#) | [DOI:10.1103/PhysRevLett.129.032001](#) | 2021-11-03

419. Search for low-mass dilepton resonances in Higgs boson decays to four-lepton final states in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, A. et al.

[Eur.Phys.J.C 82 \(2022\) 290](#) | [arXiv:2111.01299](#) | [DOI:10.1140/epjc/s10052-022-10127-0](#) | 2021-11-01

420. Search for long-lived particles produced in association with a Z boson in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 03 \(2022\) 160](#) | [arXiv:2110.13218](#) | [DOI:10.1007/JHEP03\(2022\)160](#) | 2021-10-25

421. Measurement of the inclusive and differential WZ production cross sections, polarization angles, and triple gauge couplings in pp collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 07 \(2022\) 032](#) | [arXiv:2110.11231](#) | [DOI:10.1007/JHEP07\(2022\)032](#) | 2021-10-21

422. First Search for Exclusive Diphoton Production at High Mass with Tagged Protons in Proton-Proton Collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 129 \(2022\) 011801](#) | [arXiv:2110.05916](#) | [DOI:10.1103/PhysRevLett.129.011801](#) | 2021-10-12

423. Search for long-lived particles decaying to leptons with large impact parameter in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 153](#) | [arXiv:2110.04809](#) | [DOI:10.1140/epjc/s10052-022-10027-3](#) | 2021-10-10

424. Analysis of the CP structure of the Yukawa coupling between the Higgs boson and  $\tau$  leptons in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 06 \(2022\) 012](#) | [arXiv:2110.04836](#) | [DOI:10.1007/JHEP06\(2022\)012](#) | 2021-10-10

425. Measurement of double-parton scattering in inclusive production of four jets with low transverse momentum in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 01 \(2022\) 177](#) | [arXiv:2109.13822](#) | [DOI:10.1007/JHEP01\(2022\)177](#) | 2021-09-28

426. Search for heavy resonances decaying to  $Z(\nu\bar{\nu})V(q\bar{q}')$  in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[Phys.Rev.D 106 \(2022\) 012004](#) | [arXiv:2109.08268](#) | [DOI:10.1103/PhysRevD.106.012004](#) | 2021-09-17

427. Search for heavy resonances decaying to WW, WZ, or WH boson pairs in the lepton plus merged jet final state in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 032008](#) | [arXiv:2109.06055](#) | [DOI:10.1103/PhysRevD.105.032008](#) | 2021-09-13

428. Study of quark and gluon jet substructure in Z+jet and dijet events from pp collisions

Tumasyan, Armen et al.

[JHEP 01 \(2022\) 188](#) | [arXiv:2109.03340](#) | [DOI:10.1007/JHEP01\(2022\)188](#) | 2021-09-07

429. Observation of  $B^0_s$  mesons and measurement of the  $B^0_s/B^+$  yield ratio in PbPb collisions at  $\sqrt{s_{NN}} = 5.02$  TeV

Tumasyan, Armen et al.

[Phys.Lett.B 829 \(2022\) 137062](#) | [arXiv:2109.01908](#) | [DOI:10.1016/j.physletb.2022.137062](#) | 2021-09-04

430. Observation of  $tW$  production in the single-lepton channel in pp collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 111](#) | [arXiv:2109.01706](#) | [DOI:10.1007/JHEP11\(2021\)111](#) | 2021-09-03

431. Measurement of the top quark mass using events with a single reconstructed top quark in pp collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 12 \(2021\) 161](#) | [arXiv:2108.10407](#) | [DOI:10.1007/JHEP12\(2021\)161](#) | 2021-08-23

432. Measurement of differential  $t\bar{t}$  production cross sections in the full kinematic range using lepton+jets events from proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[Phys.Rev.D 104 \(2021\) 092013](#) | [arXiv:2108.02803](#) | [DOI:10.1103/PhysRevD.104.092013](#) | 2021-08-05

433. Probing effective field theory operators in the associated production of top quarks with a Z boson in multilepton final states at  $\sqrt{s} = 13$  TeV

Lee, Kyeongpil et al.



[JHEP 12 \(2021\) 083](#) | [arXiv:2107.13896](#) | [DOI:10.1007/JHEP12\(2021\)083](#) | 2021-07-29

434. Search for new particles in events with energetic jets and large missing transverse momentum in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 153](#) | [arXiv:2107.13021](#) | [DOI:10.1007/JHEP11\(2021\)153](#) | 2021-07-27

435. Search for chargino-neutralino production in events with Higgs and W bosons using  $137 \text{ fb}^{-1}$  of proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 045](#) | [arXiv:2107.12553](#) | [DOI:10.1007/JHEP10\(2021\)045](#) | 2021-07-27

436. Measurement of the inclusive and differential Higgs boson production cross sections in the decay mode to a pair of  $\tau$  leptons in pp collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 128 \(2022\) 081805](#) | [arXiv:2107.11486](#) | [DOI:10.1103/PhysRevLett.128.081805](#) | 2021-07-23

437. Combined searches for the production of supersymmetric top quark partners in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 81 \(2021\) 970](#) | [arXiv:2107.10892](#) | [DOI:10.1140/epjc/s10052-021-09721-5](#) | 2021-07-22

438. Search for Long-Lived Particles Decaying in the CMS End Cap Muon Detectors in Proton-Proton Collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 127 \(2021\) 261804](#) | [arXiv:2107.04838](#) | [DOI:10.1103/PhysRevLett.127.261804](#) | 2021-07-10

439. Measurement of the inclusive and differential  $t\bar{t}\gamma$  cross sections in the single-lepton channel and EFT interpretation at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 12 \(2021\) 180](#) | [arXiv:2107.01508](#) | [DOI:10.1007/JHEP12\(2021\)180](#) | 2021-07-03

440. Measurement of prompt open-charm production cross sections in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 225](#) | [arXiv:2107.01476](#) | [DOI:10.1007/JHEP11\(2021\)225](#) | 2021-07-03

441. Measurements of the electroweak diboson production cross sections in proton-proton collisions at  $\sqrt{s} = 5.02$  TeV using leptonic decays

Tumasyan, Armen et al.

[Phys.Rev.Lett. 127 \(2021\) 191801](#) | [arXiv:2107.01137](#) | [DOI:10.1103/PhysRevLett.127.191801](#) | 2021-07-02

442. Search for electroweak production of charginos and neutralinos in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 147](#) | [arXiv:2106.14246](#) | [DOI:10.1007/JHEP04\(2022\)147](#) | 2021-06-27

443. Fragmentation of jets containing a prompt  $J/\psi$  meson in PbPb and pp collisions at  $\sqrt{s_{NN}} = 5.02$  TeV

Tumasyan, Armen et al.

[Phys.Lett.B 825 \(2022\) 136842](#) | [arXiv:2106.13235](#) | [DOI:10.1016/j.physletb.2021.136842](#) | 2021-06-23

444. Measurement of the electroweak production of  $Z\gamma$  and two jets in proton-proton collisions at  $\sqrt{s} = 13$  TeV and constraints on anomalous quartic gauge couplings

Tumasyan, Armen et al.

[Phys.Rev.D 104 \(2021\) 072001](#) | [arXiv:2106.11082](#) | [DOI:10.1103/PhysRevD.104.072001](#) | 2021-06-21

445. Search for  $W\gamma$  resonances in proton-proton collisions at  $\sqrt{s} = 13$  TeV using hadronic decays of Lorentz-boosted W bosons

Tumasyan, Armen et al.

[Phys.Lett.B 826 \(2022\) 136888](#) | [arXiv:2106.10509](#) | [DOI:10.1016/j.physletb.2022.136888](#) | 2021-06-19

446. Search for a heavy Higgs boson decaying into two lighter Higgs bosons in the  $\tau\tau b\bar{b}$  final state at 13 TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 057](#) | [arXiv:2106.10361](#) | [DOI:10.1007/JHEP11\(2021\)057](#) | 2021-06-18

447. Study of Z boson plus jets events using variables sensitive to double-parton scattering in pp collisions at 13 TeV

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 176](#) | [arXiv:2105.14511](#) | [DOI:10.1007/JHEP10\(2021\)176](#) | 2021-05-30

448. Measurements of the  $pp \rightarrow W^\pm \gamma \gamma$  and  $pp \rightarrow Z \gamma \gamma$  cross sections at  $\sqrt{s} = 13$  TeV and limits on anomalous quartic gauge couplings

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 174](#) | [arXiv:2105.12780](#) | [DOI:10.1007/JHEP10\(2021\)174](#) | 2021-05-26

449. Search for strongly interacting massive particles generating trackless jets in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 213](#) | [arXiv:2105.09178](#) | [DOI:10.1140/epjc/s10052-022-10095-5](#) | 2021-05-19

450. Search for lepton-flavor violating decays of the Higgs boson in the  $\mu\tau$  and  $e\tau$  final states in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 032013](#) | [arXiv:2105.03007](#) | [DOI:10.1103/PhysRevD.104.032013](#) | 2021-05-06

451. Search for long-lived particles decaying to jets with displaced vertices in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, A.M. et al.

[Phys.Rev.D 104 \(2021\) 052011](#) | [arXiv:2104.13474](#) | [DOI:10.1103/PhysRevD.104.052011](#) | 2021-04-27

452. Search for a heavy resonance decaying to a top quark and a W boson at  $\sqrt{s} = 13$  TeV in the fully hadronic final state

Sirunyan, Albert M et al.

[JHEP 12 \(2021\) 106](#) | [arXiv:2104.12853](#) | [DOI:10.1007/JHEP12\(2021\)106](#) | 2021-04-26

453. Constraints on anomalous Higgs boson couplings to vector bosons and fermions in its production and decay using the four-lepton final state

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 052004](#) | [arXiv:2104.12152](#) | [DOI:10.1103/PhysRevD.104.052004](#) | 2021-04-25

454. Search for  $W'$  bosons decaying to a top and a bottom quark at  $\sqrt{s} = 13$  TeV in the hadronic final state

Sirunyan, Albert M et al.

[Phys.Lett.B 820 \(2021\) 136535](#) | [arXiv:2104.04831](#) | [DOI:10.1016/j.physletb.2021.136535](#) | 2021-04-10

455. Search for charged Higgs bosons produced in vector boson fusion processes and decaying into vector boson pairs in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 723](#) | [arXiv:2104.04762](#) | [DOI:10.1140/epjc/s10052-021-09472-3](#) | 2021-04-10

456. Precision luminosity measurement in proton-proton collisions at  $\sqrt{s} = 13$  TeV in 2015 and 2016 at CMS

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 800](#) | [arXiv:2104.01927](#) | [DOI:10.1140/epjc/s10052-021-09538-2](#) | 2021-04-05

457. Constraints on the Initial State of Pb-Pb Collisions via Measurements of Z-Boson Yields and Azimuthal Anisotropy at  $\sqrt{s_{NN}}=5.02$  TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 127 \(2021\) 102002](#) | [arXiv:2103.14089](#) | [DOI:10.1103/PhysRevLett.127.102002](#) | 2021-03-25

458. Measurements of Higgs boson production cross sections and couplings in the diphoton decay channel at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2021\) 027](#) | [arXiv:2103.06956](#) | [DOI:10.1007/JHEP07\(2021\)027](#) | 2021-03-11

459. Measurements of production cross sections of the Higgs boson in the four-lepton final state in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 488](#) | [arXiv:2103.04956](#) | [DOI:10.1140/epjc/s10052-021-09200-x](#) | 2021-03-08

460. Using Z Boson Events to Study Parton-Medium Interactions in Pb-Pb Collisions

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 128 \(2022\) 122301](#) | [arXiv:2103.04377](#) | [DOI:10.1103/PhysRevLett.128.122301](#) | 2021-03-07

461. Search for resonant and nonresonant new phenomena in high-mass dilepton final states at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2021\) 208](#) | [arXiv:2103.02708](#) | [DOI:10.1007/JHEP07\(2021\)208](#) | 2021-03-03

462. Search for top squark production in fully-hadronic final states in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 052001](#) | [arXiv:2103.01290](#) | [DOI:10.1103/PhysRevD.104.052001](#) | 2021-03-01

463. Study of Drell-Yan dimuon production in proton-lead collisions at  $\sqrt{s_{NN}} = 8.16$  TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 182](#) | [arXiv:2102.13648](#) | [DOI:10.1007/JHEP05\(2021\)182](#) | 2021-02-26

464. First measurement of large area jet transverse momentum spectra in heavy-ion collisions

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 284](#) | [arXiv:2102.13080](#) | [DOI:10.1007/JHEP05\(2021\)284](#) | 2021-02-25

465. Evidence for X(3872) in Pb-Pb Collisions and Studies of its Prompt Production at  $\sqrt{s_{NN}}=5.02$  TeV

Sirunyan, Albert M. et al.

[Phys.Rev.Lett. 128 \(2022\) 032001](#) | [arXiv:2102.13048](#) | [DOI:10.1103/PhysRevLett.128.032001](#) | 2021-02-25

466. Measurements of angular distance and momentum ratio distributions in three-jet and Z + two-jet final states in pp collisions

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 852](#) | [arXiv:2102.08816](#) | [DOI:10.1140/epjc/s10052-021-09570-2](#) | 2021-02-17

467. Search for a heavy vector resonance decaying to a  $Z^0$  boson and a Higgs boson in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 688](#) | [arXiv:2102.08198](#) | [DOI:10.1140/epjc/s10052-021-09348-6](#) | 2021-02-16

468. Search for top squarks in final states with two top quarks and several light-flavor jets in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 032006](#) | [arXiv:2102.06976](#) | [DOI:10.1103/PhysRevD.104.032006](#) | 2021-02-13

469. Hard color-singlet exchange in dijet events in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 032009](#) | [arXiv:2102.06945](#) | [DOI:10.1103/PhysRevD.104.032009](#) | 2021-02-13

470. Performance of the CMS muon trigger system in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[JINST 16 \(2021\) P07001](#) | [arXiv:2102.04790](#) | [DOI:10.1088/1748-0221/16/07/P07001](#) | 2021-02-09

471. Observation of a New Excited Beauty Strange Baryon Decaying to  $\Xi^- \pi^+ \pi^-$

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 126 \(2021\) 252003](#) | [arXiv:2102.04524](#) | [DOI:10.1103/PhysRevLett.126.252003](#) | 2021-02-08

472. Measurement of the  $W\gamma$  Production Cross Section in Proton-Proton Collisions at  $\sqrt{s}=13$  TeV and Constraints on Effective Field Theory Coefficients

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 126 \(2021\) 252002](#) | [arXiv:2102.02283](#) | [DOI:10.1103/PhysRevLett.126.252002](#) | 2021-02-03

473. Measurements of the differential cross sections of the production of  $Z + \text{jets}$  and  $\gamma + \text{jets}$  and of  $Z$  boson emission collinear with a jet in pp collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 285](#) | [arXiv:2102.02238](#) | [DOI:10.1007/JHEP05\(2021\)285](#) | 2021-02-03

474. In-medium modification of dijets in PbPb collisions at  $\sqrt{s_{NN}} = 5.02$  TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 116](#) | [arXiv:2101.04720](#) | [DOI:10.1007/JHEP05\(2021\)116](#) | 2021-01-12

475. Measurement of the  $Z$  boson differential production cross section using its invisible decay mode ( $Z\nu\bar{\nu}$ ) in proton-proton collisions at  $\sqrt{s}=13$  TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 205](#) | [arXiv:2012.09254](#) | [DOI:10.1007/JHEP05\(2021\)205](#) | 2020-12-16

476. First measurement of the cross section for top quark pair production with additional charm jets using dileptonic final states in pp collisions at  $s=13\text{TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 820 \(2021\) 136565](#) | [arXiv:2012.09225](#) | [DOI:10.1016/j.physletb.2021.136565](#) | 2020-12-16

477. Search for supersymmetry in final states with two oppositely charged same-flavor leptons and missing transverse momentum in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2021\) 123](#) | [arXiv:2012.08600](#) | [DOI:10.1007/JHEP04\(2021\)123](#) | 2020-12-15

478. Electron and photon reconstruction and identification with the CMS experiment at the CERN LHC

Sirunyan, Albert M et al.

[JINST 16 \(2021\) P05014](#) | [arXiv:2012.06888](#) | [DOI:10.1088/1748-0221/16/05/P05014](#) | 2020-12-12

479. Measurement of differential cross sections for  $Z$  bosons produced in association with charm jets in pp collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2021\) 109](#) | [arXiv:2012.04119](#) | [DOI:10.1007/JHEP04\(2021\)109](#) | 2020-12-07

480. Search for new physics in top quark production with additional leptons in proton-proton collisions at  $\sqrt{s} = 13$  TeV using effective field theory

Sirunyan, Albert M et al.

481. Search for singly and pair-produced leptoquarks coupling to third-generation fermions in proton-proton collisions at  $\sqrt{s}=13\text{TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 819 \(2021\) 136446](#) | [arXiv:2012.04178](#) | [DOI:10.1016/j.physletb.2021.136446](#) | 2020-12-07

482. Search for long-lived particles using displaced jets in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 012015](#) | [arXiv:2012.01581](#) | [DOI:10.1103/PhysRevD.104.012015](#) | 2020-12-02

483. Search for nonresonant Higgs boson pair production in final states with two bottom quarks and two photons in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 03 \(2021\) 257](#) | [arXiv:2011.12373](#) | [DOI:10.1007/JHEP03\(2021\)257](#) | 2020-11-24

484. Search for the rare decay of the W boson into a pion and a photon in proton-proton collisions at  $\sqrt{s}=13\text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 819 \(2021\) 136409](#) | [arXiv:2011.06028](#) | [DOI:10.1016/j.physletb.2021.136409](#) | 2020-11-11

485. Observation of Forward Neutron Multiplicity Dependence of Dimuon Acoplanarity in Ultraperipheral Pb-Pb Collisions at  $\sqrt{s_{\text{NN}}}=5.02\text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 127 \(2021\) 122001](#) | [arXiv:2011.05239](#) | [DOI:10.1103/PhysRevLett.127.122001](#) | 2020-11-10

486. Development and validation of HERWIG 7 tunes from CMS underlying-event measurements

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 312](#) | [arXiv:2011.03422](#) | [DOI:10.1140/epjc/s10052-021-08949-5](#) | 2020-11-06

487. Measurement of the Higgs boson production rate in association with top quarks in final states with electrons, muons, and hadronically decaying tau leptons at  $\sqrt{s} = 13\text{ TeV}$

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 378](#) | [arXiv:2011.03652](#) | [DOI:10.1140/epjc/s10052-021-09014-x](#) | 2020-11-06

488. The very forward CASTOR calorimeter of the CMS experiment

Khachatryan, Vardan et al.

[JINST 16 \(2021\) P02010](#) | [arXiv:2011.01185](#) | [DOI:10.1088/1748-0221/16/02/P02010](#) | 2020-11-02

489. Angular analysis of the decay  $B^+ \rightarrow K^{*(892)^+} \mu^+ \mu^-$  in proton-proton collisions at  $\sqrt{s} = 8\text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 04 \(2021\) 124](#) | [arXiv:2010.13968](#) | [DOI:10.1007/JHEP04\(2021\)124](#) | 2020-10-26

490. MUSiC: a model-unspecific search for new physics in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 629](#) | [arXiv:2010.02984](#) | [DOI:10.1140/epjc/s10052-021-09236-z](#) | 2020-10-06

491. Search for dark photons in Higgs boson production via vector boson fusion in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 03 \(2021\) 011](#) | [arXiv:2009.14009](#) | [DOI:10.1007/JHEP03\(2021\)011](#) | 2020-09-29

492. Measurement of prompt  $D^0$  and  $\bar{D}^0$  meson azimuthal anisotropy and search for strong electric fields in PbPb collisions at  $\sqrt{s_{\text{NN}}} = 5.02\text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 816 \(2021\) 136253](#) | [arXiv:2009.12628](#) | [DOI:10.1016/j.physletb.2021.136253](#) | 2020-09-26

493. Measurements of production cross sections of polarized same-sign W boson pairs in association with two jets in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 812 \(2021\) 136018](#) | [arXiv:2009.09429](#) | [DOI:10.1016/j.physletb.2020.136018](#) | 2020-09-20

494. Measurement of the top quark Yukawa coupling from  $t\bar{t}$  kinematic distributions in the dilepton final state in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

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Sirunyan, Albert M et al.

[Phys.Lett.B 813 \(2021\) 136036](#) | [arXiv:2009.07065](#) | [DOI:10.1016/j.physletb.2020.136036](#) | 2020-09-15

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497. Measurements of  $pp \rightarrow ZZ$  production cross sections and constraints on anomalous triple gauge couplings at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 200](#) | [arXiv:2009.01186](#) | [DOI:10.1140/epjc/s10052-020-08817-8](#) | 2020-09-02

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Sirunyan, Albert M et al.

[Phys.Rev.D 102 \(2020\) 092001](#) | [arXiv:2009.00119](#) | [DOI:10.1103/PhysRevD.102.092001](#) | 2020-08-31

499. Observation of electroweak production of  $W\gamma$  with two jets in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 811 \(2020\) 135988](#) | [arXiv:2008.10521](#) | [DOI:10.1016/j.physletb.2020.135988](#) | 2020-08-24

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[Phys.Rev.D 102 \(2020\) 112004](#) | [arXiv:2008.09835](#) | [DOI:10.1103/PhysRevD.102.112004](#) | 2020-08-22

501. Measurement of  $B_c(2S)^+$  and  $B_c^{*+}(2S)^+$  cross section ratios in proton-proton collisions at  $\sqrt{s} = 13$  TeV

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[Phys.Rev.D 102 \(2020\) 092007](#) | [arXiv:2008.08629](#) | [DOI:10.1103/PhysRevD.102.092007](#) | 2020-08-19

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[Phys.Rev.D 103 \(2021\) 052008](#) | [arXiv:2008.07860](#) | [DOI:10.1103/PhysRevD.103.052008](#) | 2020-08-18

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[Phys.Lett.B 812 \(2021\) 135992](#) | [arXiv:2008.07013](#) | [DOI:10.1016/j.physletb.2020.135992](#) | 2020-08-16

504. Search for top squark pair production using dilepton final states in pp collision data collected at  $\sqrt{s} = 13$  TeV

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[Phys.Rev.D 102 \(2020\) 092012](#) | [arXiv:2008.04174](#) | [DOI:10.1103/PhysRevD.102.092012](#) | 2020-08-10

507. Search for supersymmetry in proton-proton collisions at  $\sqrt{s} = 13$  TeV in events with high-momentum Z bosons and missing transverse momentum

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[JHEP 11 \(2020\) 039](#) | [arXiv:2007.05122](#) | [DOI:10.1007/JHEP11\(2020\)039](#) | 2020-07-10

509. Search for the lepton flavor violating decay  $\tau \rightarrow 3\mu$  in proton-proton collisions at  $\sqrt{s} = 13$  TeV

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[JHEP 01 \(2021\) 163](#) | [arXiv:2007.05658](#) | [DOI:10.1007/JHEP01\(2021\)163](#) | 2020-07-10

510. Investigation into the event-activity dependence of  $Y(nS)$  relative production in proton-proton collisions at  $\sqrt{s} = 7$  TeV

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[JHEP 11 \(2020\) 001](#) | [arXiv:2007.04277](#) | [DOI:10.1007/JHEP11\(2020\)001](#) | 2020-07-09

511. Measurement of the CP-violating phase  $\phi_s$  in the  $B_s^0 \rightarrow J/\psi \phi(1020) \rightarrow \mu^+ \mu^- K^+ K^-$  channel in proton-proton collisions at  $\sqrt{s} = 13$  TeV

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[Phys.Lett.B 816 \(2021\) 136188](#) | [arXiv:2007.02434](#) | [DOI:10.1016/j.physletb.2021.136188](#) | 2020-07-05

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[JHEP 03 \(2021\) 003](#) | [arXiv:2007.01984](#) | [DOI:10.1007/JHEP03\(2021\)003](#) | 2020-07-03

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Tumasyan, Armen et al.

[JHEP 07 \(2023\) 161](#) | [arXiv:2206.09997](#) | [DOI:10.1007/JHEP07\(2023\)161](#) | 2020

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Sirunyan, Albert M et al.

[JINST 15 \(2020\) P10002](#) | [arXiv:2006.14359](#) | [DOI:10.1088/1748-0221/15/10/P10002](#) | 2020-06-25

515. Inclusive search for highly boosted Higgs bosons decaying to bottom quark-antiquark pairs in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[JHEP 12 \(2020\) 085](#) | [arXiv:2006.13251](#) | [DOI:10.1007/JHEP12\(2020\)085](#) | 2020-06-23

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[Phys.Rev.Lett. 125 \(2020\) 222001](#) | [arXiv:2006.11110](#) | [DOI:10.1103/PhysRevLett.125.222001](#) | 2020-06-19

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[JINST 15 \(2020\) P10017](#) | [arXiv:2006.10165](#) | [DOI:10.1088/1748-0221/15/10/P10017](#) | 2020-06-17

518. Measurement of the azimuthal anisotropy of  $\Upsilon(1S)$  and  $\Upsilon(2S)$  mesons in PbPb collisions at  $\sqrt{s_{NN}} = 5.02$  TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 819 \(2021\) 136385](#) | [arXiv:2006.07707](#) | [DOI:10.1016/j.physletb.2021.136385](#) | 2020-06-13

519. Search for resonant pair production of Higgs bosons in the  $bbZZ$  channel in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 102 \(2020\) 032003](#) | [arXiv:2006.06391](#) | [DOI:10.1103/PhysRevD.102.032003](#) | 2020-06-11

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Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 054](#) | [arXiv:2005.14219](#) | [DOI:10.1007/JHEP05\(2021\)054](#) | 2020-05-28

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Sirunyan, Albert M et al.

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Sirunyan, Albert M et al.

[Phys.Rev.D 102 \(2020\) 072001](#) | [arXiv:2005.08900](#) | [DOI:10.1103/PhysRevD.102.072001](#) | 2020-05-18

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Sirunyan, Albert M et al.

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[Phys.Rev.Lett. 125 \(2020\) 152001](#) | [arXiv:2005.04764](#) | [DOI:10.1103/PhysRevLett.125.152001](#) | 2020-05-10

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Aad, Georges et al.

[JHEP 08 \(2020\) 051](#) | [arXiv:2005.03799](#) | [DOI:10.1007/JHEP08\(2020\)051](#) | 2020-05-07

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[Phys.Lett.B 809 \(2020\) 135710](#) | [arXiv:2005.01173](#) | [DOI:10.1016/j.physletb.2020.135710](#) | 2020-05-03

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[Phys.Lett.B 808 \(2020\) 135609](#) | [arXiv:2004.12181](#) | [DOI:10.1016/j.physletb.2020.135609](#) | 2020-04-25

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[JINST 15 \(2020\) P06005](#) | [arXiv:2004.08262](#) | [DOI:10.1088/1748-0221/15/06/P06005](#) | 2020-04-17

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Sirunyan, Albert M et al.

[Phys.Lett.B 806 \(2020\) 135502](#) | [arXiv:2004.05153](#) | [DOI:10.1016/j.physletb.2020.135502](#) | 2020-04-10

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Sirunyan, Albert M et al.

[JHEP 07 \(2020\) 115](#) | [arXiv:2004.00602](#) | [DOI:10.1007/JHEP07\(2020\)115](#) | 2020-04-01

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Sirunyan, Albert M et al.

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532. Search for supersymmetry in pp collisions at  $\sqrt{s} = 13$  TeV with  $137 \text{ fb}^{-1}$  in final states with a single lepton using the sum of masses of large-radius jets

Sirunyan, Albert M et al.

[Phys.Rev.D 101 \(2020\) 052010](#) | [arXiv:1911.07558](#) | [DOI:10.1103/PhysRevD.101.052010](#) | 2020-03-25

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Sirunyan, Albert M et al.

[Phys.Rev.Lett. 125 \(2020\) 061801](#) | [arXiv:2003.10866](#) | [DOI:10.1103/PhysRevLett.125.061801](#) | 2020-03-24

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Sirunyan, Albert M et al.

[JHEP 07 \(2020\) 125](#) | [arXiv:2003.06467](#) | [DOI:10.1007/JHEP07\(2020\)125](#) | 2020-03-13

535. Study of central exclusive  $\pi^+ \pi^-$  production in proton-proton collisions at  $\sqrt{s} = 5.02$  and 13 TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 718](#) | [arXiv:2003.02811](#) | [DOI:10.1140/epjc/s10052-020-8166-5](#) | 2020-03-05

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Sirunyan, Albert M et al.

[JINST 15 \(2020\) P09018](#) | [arXiv:2003.00503](#) | [DOI:10.1088/1748-0221/15/09/P09018](#) | 2020-03-01

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Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 1164](#) | [arXiv:2002.12146](#) | [DOI:10.1140/epjc/s10052-020-08562-y](#) | 2020-02-27

538. Measurement of the cross section for electroweak production of a Z boson, a photon and two jets in proton-proton collisions at  $\sqrt{s} = 13$  TeV and constraints on anomalous quartic couplings

Sirunyan, Albert M et al.

[JHEP 06 \(2020\) 076](#) | [arXiv:2002.09902](#) | [DOI:10.1007/JHEP06\(2020\)076](#) | 2020-02-23

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[Phys.Lett.B 808 \(2020\) 135578](#) | [arXiv:2002.06393](#) | [DOI:10.1016/j.physletb.2020.135578](#) | 2020-02-15

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Sirunyan, Albert M et al.

[Phys.Lett.B 805 \(2020\) 135425](#) | [arXiv:2002.06398](#) | [DOI:10.1016/j.physletb.2020.135425](#) | 2020-02-15

541. Search for physics beyond the standard model in events with jets and two same-sign or at least three charged leptons in proton-proton collisions at  $\sqrt{s}=13$  TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 752](#) | [arXiv:2001.10086](#) | [DOI:10.1140/epjc/s10052-020-8168-3](#) | 2020-01-27

542. Search for charged Higgs bosons decaying into a top and a bottom quark in the all-jet final state of pp collisions at  $\sqrt{s}=13$  TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2020\) 126](#) | [arXiv:2001.07763](#) | [DOI:10.1007/JHEP07\(2020\)126](#) | 2020-01-21

543. Measurement of the associated production of a Z boson with charm or bottom quark jets in proton-proton collisions at  $\sqrt{s}=13$  TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 102 \(2020\) 032007](#) | [arXiv:2001.06899](#) | [DOI:10.1103/PhysRevD.102.032007](#) | 2020-01-19

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Sirunyan, Albert M et al.

[JINST 15 \(2020\) P06009](#) | [arXiv:2001.06553](#) | [DOI:10.1088/1748-0221/15/06/P06009](#) | 2020-01-17

545. Study of excited  $\Lambda_b^0$  states decaying to  $\Lambda_b^0\pi^+\pi^-$  in proton-proton collisions at  $\sqrt{s}=13$  TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 803 \(2020\) 135345](#) | [arXiv:2001.06533](#) | [DOI:10.1016/j.physletb.2020.135345](#) | 2020-01-17

546. Search for an excited lepton that decays via a contact interaction to a lepton and two jets in proton-proton collisions at  $\sqrt{s}=13$  TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2020\) 052](#) | [arXiv:2001.04521](#) | [DOI:10.1007/JHEP05\(2020\)052](#) | 2020-01-13

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Sirunyan, Albert M et al.

[Mach.Learn.Sci.Tech. 1 \(2020\) 035012](#) | [arXiv:1912.12238](#) | [DOI:10.1088/2632-2153/ab9023](#) | 2019-12-27

548. Measurement of the top quark forward-backward production asymmetry and the anomalous chromoelectric and chromomagnetic moments in pp collisions at  $\sqrt{s}=13$  TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2020\) 146](#) | [arXiv:1912.09540](#) | [DOI:10.1007/JHEP06\(2020\)146](#) | 2019-12-19

549. Search for direct top squark pair production in events with one lepton, jets, and missing transverse momentum at 13 TeV with the CMS experiment

Sirunyan, Albert M et al.

[JHEP 05 \(2020\) 032](#) | [arXiv:1912.08887](#) | [DOI:10.1007/JHEP05\(2020\)032](#) | 2019-12-18

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[Phys.Rev.Lett. 124 \(2020\) 162002](#) | [arXiv:1912.07706](#) | [DOI:10.1103/PhysRevLett.124.162002](#) | 2019-12-16

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[Comput.Softw.Big Sci. 4 \(2020\) 10](#) | [arXiv:1912.06046](#) | [DOI:10.1007/s41781-020-00041-z](#) | 2019-12-12

552. Search for a Narrow Resonance Lighter than 200 GeV Decaying to a Pair of Muons in Proton-Proton Collisions at  $\sqrt{s}=13$  TeV

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[Phys.Rev.Lett. 124 \(2020\) 131802](#) | [arXiv:1912.04776](#) | [DOI:10.1103/PhysRevLett.124.131802](#) | 2019-12-10

553. Determination of the strong coupling constant  $\alpha_S(m_Z)$  from measurements of inclusive  $W^\pm$  and Z boson production cross sections in proton-proton collisions at  $\sqrt{s} = 7$  and 8 TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2020\) 018](#) | [arXiv:1912.04387](#) | [DOI:10.1007/JHEP06\(2020\)018](#) | 2019-12-09

554. Performance of the reconstruction and identification of high-momentum muons in proton-proton collisions at  $\sqrt{s} = 13$  TeV

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[JINST 15 \(2020\) P02027](#) | [arXiv:1912.03516](#) | [DOI:10.1088/1748-0221/15/02/P02027](#) | 2019-12-07

555. Search for a heavy Higgs boson decaying to a pair of W bosons in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 034](#) | [arXiv:1912.01594](#) | [DOI:10.1007/JHEP03\(2020\)034](#) | 2019-12-03

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Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 131](#) | [arXiv:1912.01662](#) | [DOI:10.1007/JHEP03\(2020\)131](#) | 2019-12-03

557. Measurement of the top quark pair production cross section in dilepton final states containing one  $\tau$  lepton in pp collisions at  $\sqrt{s} = 13$  TeV

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[JHEP 02 \(2020\) 191](#) | [arXiv:1911.13204](#) | [DOI:10.1007/JHEP02\(2020\)191](#) | 2019-11-29

558. Search for lepton flavour violating decays of a neutral heavy Higgs boson to  $\mu\tau$  and  $e\tau$  in proton-proton collisions at  $\sqrt{s} = 13$  TeV

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[JHEP 03 \(2020\) 103](#) | [arXiv:1911.10267](#) | [DOI:10.1007/JHEP03\(2020\)103](#) | 2019-11-22

559. Search for physics beyond the standard model in multilepton final states in proton-proton collisions at  $\sqrt{s} = 13$  TeV

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[JHEP 03 \(2020\) 051](#) | [arXiv:1911.04968](#) | [DOI:10.1007/JHEP03\(2020\)051](#) | 2019-11-12

560. Search for high mass dijet resonances with a new background prediction method in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2020\) 033](#) | [arXiv:1911.03947](#) | [DOI:10.1007/JHEP05\(2020\)033](#) | 2019-11-10

561. Measurement of the Jet Mass Distribution and Top Quark Mass in Hadronic Decays of Boosted Top Quarks in pp Collisions at  $\sqrt{s} = 13$  TeV

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[Phys.Rev.Lett. 124 \(2020\) 202001](#) | [arXiv:1911.03800](#) | [DOI:10.1103/PhysRevLett.124.202001](#) | 2019-11-09

562. Search for dijet resonances using events with three jets in proton-proton collisions at  $\sqrt{s} = 13$  TeV

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[Phys.Lett.B 805 \(2020\) 135448](#) | [arXiv:1911.03761](#) | [DOI:10.1016/j.physletb.2020.135448](#) | 2019-11-09

563. Observation of the  $\Lambda_b^0 \rightarrow J/\psi \Lambda \phi$  decay in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 802 \(2020\) 135203](#) | [arXiv:1911.03789](#) | [DOI:10.1016/j.physletb.2020.135203](#) | 2019-11-09

564. Search for new neutral Higgs bosons through the  $H \rightarrow ZA \rightarrow \ell^+ \ell^- b\bar{b}$  process in pp collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 055](#) | [arXiv:1911.03781](#) | [DOI:10.1007/JHEP03\(2020\)055](#) | 2019-11-09

565. Studies of charm quark diffusion inside jets using PbPb and pp collisions at  $\sqrt{s_{\text{NN}}} = 5.02$  TeV

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[Phys.Rev.Lett. 125 \(2020\) 102001](#) | [arXiv:1911.01461](#) | [DOI:10.1103/PhysRevLett.125.102001](#) | 2019-11-04

566. Search for top squark pair production in a final state with two tau leptons in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[JHEP 02 \(2020\) 015](#) | [arXiv:1910.12932](#) | [DOI:10.1007/JHEP02\(2020\)015](#) | 2019-10-28

567. Measurement of properties of  $B^0_s \rightarrow \mu^+ \mu^-$  decays and search for  $B^0 \rightarrow \mu^+ \mu^-$  with the CMS experiment

Sirunyan, Albert M et al.

[JHEP 04 \(2020\) 188](#) | [arXiv:1910.12127](#) | [DOI:10.1007/JHEP04\(2020\)188](#) | 2019-10-26

568. Search for a heavy pseudoscalar Higgs boson decaying into a 125 GeV Higgs boson and a Z boson in final states with two tau and two light leptons at  $\sqrt{s} = 13$  TeV

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[JHEP 03 \(2020\) 065](#) | [arXiv:1910.11634](#) | [DOI:10.1007/JHEP03\(2020\)065](#) | 2019-10-25

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Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 014](#) | [arXiv:1910.08815](#) | [DOI:10.1007/JHEP03\(2020\)014](#) | 2019-10-19

570. Mixed higher-order anisotropic flow and nonlinear response coefficients of charged particles in PbPb collisions at  $\sqrt{\langle b \rangle s_{\text{NN}}} = 2.76$  and 5.02 TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 534](#) | [arXiv:1910.08789](#) | [DOI:10.1140/epjc/s10052-020-7834-9](#) | 2019-10-19

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Sirunyan, Albert M et al.

[Phys.Rev.C 101 \(2020\) 064906](#) | [arXiv:1910.04812](#) | [DOI:10.1103/PhysRevC.101.064906](#) | 2019-10-10

572. Measurement of the top quark polarization and  $t\bar{t}$  spin correlations using dilepton final states in proton-proton collisions at  $\sqrt{s} = 13$  TeV

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[Phys.Rev.D 100 \(2019\) 072002](#) | [arXiv:1907.03729](#) | [DOI:10.1103/PhysRevD.100.072002](#) | 2019-10-09

573. Study of  $J/\psi$  meson production inside jets in pp collisions at  $\sqrt{s} = 8$  TeV

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574. Search for Supersymmetry with a Compressed Mass Spectrum in Events with a Soft  $\tau$  Lepton, a Highly Energetic Jet, and Large Missing Transverse Momentum in Proton-Proton Collisions at  $\sqrt{s} = 13$  TeV

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[JINST 15 \(2020\) P05002](#) | [arXiv:1910.00079](#) | [DOI:10.1088/1748-0221/15/05/P05002](#) | 2019-09-30

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[Phys.Lett.B 803 \(2020\) 135285](#) | [arXiv:1909.05306](#) | [DOI:10.1016/j.physletb.2020.135285](#) | 2019-09-11

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[JHEP 01 \(2020\) 036](#) | [arXiv:1909.04721](#) | [DOI:10.1007/JHEP01\(2020\)036](#) | 2019-09-10

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[JHEP 12 \(2019\) 061](#) | [arXiv:1909.04133](#) | [DOI:10.1007/JHEP12\(2019\)061](#) | 2019-09-09

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[Eur.Phys.J.C 79 \(2019\) 969](#) | [arXiv:1907.08155](#) | [DOI:10.1140/epjc/s10052-019-7451-7](#) | 2019-07-18

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[Phys.Rev.Lett. 123 \(2019\) 241801](#) | [arXiv:1907.06275](#) | [DOI:10.1103/PhysRevLett.123.241801](#) | 2019-07-14

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[Phys.Rev.D 100 \(2019\) 072001](#) | [arXiv:1906.11903](#) | [DOI:10.1103/PhysRevD.100.072001](#) | 2019-06-27

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[Phys.Lett.B 797 \(2019\) 134876](#) | [arXiv:1906.06441](#) | [DOI:10.1016/j.physletb.2019.134876](#) | 2019-06-14

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[Eur.Phys.J.C 80 \(2020\) 237](#) | [arXiv:1906.05977](#) | [DOI:10.1140/epjc/s10052-020-7773-5](#) | 2019-06-13

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[Phys.Lett.B 803 \(2020\) 135328](#) | [arXiv:1906.03322](#) | [DOI:10.1016/j.physletb.2020.135328](#) | 2019-06-07

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[Phys.Lett.B 800 \(2020\) 135048](#) | [arXiv:1905.01486](#) | [DOI:10.1016/j.physletb.2019.135048](#) | 2019-05-04

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[Eur.Phys.J.C 79 \(2019\) 886](#) | [arXiv:1903.11144](#) | [DOI:10.1140/epjc/s10052-019-7387-y](#) | 2019-03-26

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625. Search for charged Higgs bosons in the  $H^{\pm} \rightarrow \tau^{\pm} \nu_{\tau}$  decay channel in proton-proton collisions at  $\sqrt{s} = 13$  TeV

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[Eur.Phys.J.C 80 \(2020\) 43](#) | [arXiv:1903.04040](#) | [DOI:10.1140/epjc/s10052-019-7585-7](#) | 2019-03-10

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[Eur.Phys.J.C 79 \(2019\) 773](#) | [arXiv:1902.04374](#) | [DOI:10.1140/epjc/s10052-019-7276-4](#) | 2019-02-12

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[Phys.Lett.B 799 \(2019\) 135049](#) | [arXiv:1902.03603](#) | [DOI:10.1016/j.physletb.2019.135049](#) | 2019-02-10

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[Eur.Phys.J.C 79 \(2019\) 702](#) | [arXiv:1902.01339](#) | [DOI:10.1140/epjc/s10052-019-7202-9](#) | 2019-02-04

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[Phys.Rev.Lett. 122 \(2019\) 151802](#) | [arXiv:1901.11201](#) | [DOI:10.1103/PhysRevLett.122.151802](#) | 2019-01-30

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637. Measurement of electroweak WZ boson production and search for new physics in WZ + two jets events in pp collisions at  $\sqrt{s} = 13$  TeV

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[Phys.Rev.D 99 \(2019\) 112003](#) | [arXiv:1901.00174](#) | [DOI:10.1103/PhysRevD.99.112003](#) | 2019-01-01

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[JHEP 04 \(2019\) 114](#) | [arXiv:1812.10443](#) | [DOI:10.1007/JHEP04\(2019\)114](#) | 2018-12-26

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[Eur.Phys.J.C 79 \(2019\) 313](#) | [arXiv:1812.10534](#) | [DOI:10.1140/epjc/s10052-019-6788-2](#) | 2018-12-26

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[Phys.Lett.B 800 \(2020\) 135042](#) | [arXiv:1812.10514](#) | [DOI:10.1016/j.physletb.2019.135042](#) | 2018-12-26

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[Eur.Phys.J.C 79 \(2019\) 364](#) | [arXiv:1812.09768](#) | [DOI:10.1140/epjc/s10052-019-6855-8](#) | 2018-12-23

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[Phys.Lett.B 792 \(2019\) 369](#) | [arXiv:1812.06504](#) | [DOI:10.1016/j.physletb.2019.03.059](#) | 2018-12-16

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[Eur.Phys.J.C 79 \(2019\) 208](#) | [arXiv:1812.06489](#) | [DOI:10.1140/epjc/s10052-019-6688-5](#) | 2018-12-16

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[Phys.Lett.B 795 \(2019\) 398](#) | [arXiv:1812.06359](#) | [DOI:10.1016/j.physletb.2019.06.021](#) | 2018-12-15

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[JHEP 03 \(2019\) 031](#) | [arXiv:1812.06302](#) | [DOI:10.1007/JHEP03\(2019\)031](#) | 2018-12-15

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654. Measurement of the energy density as a function of pseudorapidity in proton-proton collisions at  $\sqrt{s}=13$  TeV

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[Eur.Phys.J.C 79 \(2019\) 391](#) | [arXiv:1812.04095](#) | [DOI:10.1140/epjc/s10052-019-6861-x](#) | 2018-12-10

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[Phys.Lett.B 796 \(2019\) 131](#) | [arXiv:1812.00380](#) | [DOI:10.1016/j.physletb.2019.07.013](#) | 2018-12-02

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[Eur.Phys.J.C 79 \(2019\) 269](#) | [arXiv:1811.10021](#) | [DOI:10.1140/epjc/s10052-019-6752-1](#) | 2018-11-25

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[Phys.Lett.B 795 \(2019\) 76](#) | [arXiv:1811.10151](#) | [DOI:10.1016/j.physletb.2019.05.046](#) | 2018-11-25

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[Phys.Rev.D 99 \(2019\) 092005](#) | [arXiv:1811.09696](#) | [DOI:10.1103/PhysRevD.99.092005](#) | 2018-11-23

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[Eur.Phys.J.C 79 \(2019\) 305](#) | [arXiv:1811.09760](#) | [DOI:10.1140/epjc/s10052-019-6800-x](#) | 2018-11-23

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[Phys.Lett.B 793 \(2019\) 320](#) | [arXiv:1811.08459](#) | [DOI:10.1016/j.physletb.2019.03.064](#) | 2018-11-20

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[Phys.Rev.D 99 \(2019\) 032011](#) | [arXiv:1811.07991](#) | [DOI:10.1103/PhysRevD.99.032011](#) | 2018-11-19

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[JHEP 02 \(2019\) 149](#) | [arXiv:1811.06625](#) | [DOI:10.1007/JHEP02\(2019\)149](#) | 2018-11-15

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[Phys.Rev.D 99 \(2019\) 012005](#) | [arXiv:1810.11822](#) | [DOI:10.1103/PhysRevD.99.012005](#) | 2018-10-28

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[Phys.Rev.Lett. 123 \(2019\) 022001](#) | [arXiv:1810.11102](#) | [DOI:10.1103/PhysRevLett.123.022001](#) | 2018-10-25

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[JHEP 02 \(2019\) 179](#) | [arXiv:1810.10069](#) | [DOI:10.1007/JHEP02\(2019\)179](#) | 2018-10-23

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[Phys.Rev.C 100 \(2019\) 024902](#) | [arXiv:1810.05745](#) | [DOI:10.1103/PhysRevC.100.024902](#) | 2018-10-12

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[Phys.Lett.B 797 \(2019\) 134826](#) | [arXiv:1810.04602](#) | [DOI:10.1016/j.physletb.2019.134826](#) | 2018-10-10

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[Phys.Lett.B 796 \(2019\) 168](#) | [arXiv:1810.03022](#) | [DOI:10.1016/j.physletb.2019.07.014](#) | 2018-10-06

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[Phys.Lett.B 791 \(2019\) 172](#) | [arXiv:1810.01473](#) | [DOI:10.1016/j.physletb.2019.02.018](#) | 2018-10-02

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[Eur.Phys.J.C 79 \(2019\) 277](#) | [arXiv:1809.11080](#) | [DOI:10.1140/epjc/s10052-019-6774-8](#) | 2018-09-28

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[Eur.Phys.J.C 79 \(2019\) 421](#) | [arXiv:1809.10733](#) | [DOI:10.1140/epjc/s10052-019-6909-y](#) | 2018-09-27

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[Eur.Phys.J.C 79 \(2019\) 90](#) | [arXiv:1809.08597](#) | [DOI:10.1140/epjc/s10052-019-6556-3](#) | 2018-09-23

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[Phys.Rev.Lett. 121 \(2018\) 241802](#) | [arXiv:1809.05558](#) | [DOI:10.1103/PhysRevLett.121.241802](#) | 2018-09-14

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Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 939](#) | [arXiv:1809.03578](#) | [DOI:10.1140/epjc/s10052-018-6390-z](#) | 2018-09-10

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Sirunyan, Albert M et al.

[JHEP 06 \(2019\) 093](#) | [arXiv:1809.03590](#) | [DOI:10.1007/JHEP06\(2019\)093](#) | 2018-09-10

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[JINST 13 \(2018\) P10005](#) | [arXiv:1809.02816](#) | [DOI:10.1088/1748-0221/13/10/P10005](#) | 2018-09-08

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Sirunyan, Albert M et al.  
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[Phys.Rev.Lett. 121 \(2018\) 121801](#) | [arXiv:1808.08242](#) | [DOI:10.1103/PhysRevLett.121.121801](#) | 2018-08-24
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703. Search for production of Higgs boson pairs in the four b quark final state using large-area jets in proton-proton collisions at  $\sqrt{s} = 13$  TeV  
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[Phys.Rev.Lett. 122 \(2019\) 081804](#) | [arXiv:1808.01257](#) | [DOI:10.1103/PhysRevLett.122.081804](#) | 2018-08-03
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[JHEP 11 \(2018\) 079](#) | [arXiv:1807.07799](#) | [DOI:10.1007/JHEP11\(2018\)079](#) | 2018-07-20

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[JINST 13 \(2018\) P10034](#) | [arXiv:1807.03289](#) | [DOI:10.1088/1748-0221/13/10/P10034](#) | 2018-07-09

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[Eur.Phys.J.C 79 \(2019\) 123](#) | [arXiv:1807.02810](#) | [DOI:10.1140/epjc/s10052-019-6620-z](#) | 2018-07-08

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[Phys.Lett.B 789 \(2019\) 19](#) | [arXiv:1806.11073](#) | [DOI:10.1016/j.physletb.2018.11.007](#) | 2018-06-28

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[Phys.Rev.Lett. 121 \(2018\) 141801](#) | [arXiv:1806.04213](#) | [DOI:10.1103/PhysRevLett.121.141801](#) | 2018-06-11

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[Phys.Rev.Lett. 121 \(2018\) 141802](#) | [arXiv:1806.01058](#) | [DOI:10.1103/PhysRevLett.121.141802](#) | 2018-06-04

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Eur.Phys.J.C 78 (2018) 701 | [arXiv:1806.00863](#) | [DOI:10.1140/epjc/s10052-018-6148-7](#) | 2018-06-03

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Sirunyan, A. M. et al.

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Sirunyan, Albert M et al.

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735. Search for an exotic decay of the Higgs boson to a pair of light pseudoscalars in the final state with two  $b$  quarks and two  $\tau$  leptons in proton-proton collisions at  $\sqrt{s}=13$  TeV

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736. Measurement of nuclear modification factors of  $\Upsilon(1S)$ ,  $\Upsilon(2S)$ , and  $\Upsilon(3S)$  mesons in PbPb collisions at  $\sqrt{s_{NN}} = 5.02$  TeV

Sirunyan, Albert M et al.

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Sirunyan, Albert M et al.

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Sirunyan, A.M. et al.

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Sirunyan, Albert M et al.

[JHEP 10 \(2018\) 161](#) | [arXiv:1805.05145](#) | [DOI:10.1007/JHEP10\(2018\)161](#) | 2018-05-14

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Sirunyan, Albert M et al.

[JHEP 11 \(2018\) 018](#) | [arXiv:1805.04865](#) | [DOI:10.1007/JHEP11\(2018\)018](#) | 2018-05-13

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[Phys.Rev.Lett. 121 \(2018\) 062002](#) | [arXiv:1805.04736](#) | [DOI:10.1103/PhysRevLett.121.062002](#) | 2018-05-12

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[JHEP 08 \(2018\) 177](#) | [arXiv:1805.04758](#) | [DOI:10.1007/JHEP08\(2018\)177](#) | 2018-05-12

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Sirunyan, Albert M et al.

[Phys.Lett.B 790 \(2019\) 509](#) | [arXiv:1805.02248](#) | [DOI:10.1016/j.physletb.2019.01.058](#) | 2018-05-06

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Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 891](#) | [arXiv:1805.01428](#) | [DOI:10.1140/epjc/s10052-018-6332-9](#) | 2018-05-03

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[Phys.Rev.Lett. 121 \(2018\) 082301](#) | [arXiv:1804.09767](#) | [DOI:10.1103/PhysRevLett.121.082301](#) | 2018-04-25

747. Search for disappearing tracks as a signature of new long-lived particles in proton-proton collisions at  $\sqrt{s} = 13$  TeV

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[JHEP 08 \(2018\) 016](#) | [arXiv:1804.07321](#) | [DOI:10.1007/JHEP08\(2018\)016](#) | 2018-04-19

748. Measurement of differential cross sections for Z boson production in association with jets in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 965](#) | [arXiv:1804.05252](#) | [DOI:10.1140/epjc/s10052-018-6373-0](#) | 2018-04-14

749. Performance of the CMS muon detector and muon reconstruction with proton-proton collisions at  $\sqrt{s}= 13$  TeV

Sirunyan, A.M. et al.

[JINST 13 \(2018\) P06015](#) | [arXiv:1804.04528](#) | [DOI:10.1088/1748-0221/13/06/P06015](#) | 2018-04-12

750. Search for ttH production in the  $H \rightarrow b\bar{b}$  decay channel with leptonic tt decays in proton-proton collisions at  $\sqrt{s}=13$  TeV

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[JHEP 03 \(2019\) 026](#) | [arXiv:1804.03682](#) | [DOI:10.1007/JHEP03\(2019\)026](#) | 2018-04-10

751. Measurements of Higgs boson properties in the diphoton decay channel in proton-proton collisions at  $\sqrt{s} = 13$  TeV

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[JHEP 11 \(2018\) 185](#) | [arXiv:1804.02716](#) | [DOI:10.1007/JHEP11\(2018\)185](#) | 2018-04-08

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[Phys.Rev.Lett. 120 \(2018\) 231801](#) | [arXiv:1804.02610](#) | [DOI:10.1103/PhysRevLett.120.231801](#) | 2018-04-07

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Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 148](#) | [arXiv:1803.11116](#) | [DOI:10.1007/JHEP05\(2018\)148](#) | 2018-03-29

755. Search for high-mass resonances in final states with a lepton and missing transverse momentum at  $\sqrt{s}=13$  TeV

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[JHEP 06 \(2018\) 128](#) | [arXiv:1803.11133](#) | [DOI:10.1007/JHEP06\(2018\)128](#) | 2018-03-29

756. Search for a heavy resonance decaying into a Z boson and a Z or W boson in  $2\ell 2q$  final states at  $\sqrt{s}=13$  TeV

Sirunyan, Albert M et al.

[JHEP 09 \(2018\) 101](#) | [arXiv:1803.10093](#) | [DOI:10.1007/JHEP09\(2018\)101](#) | 2018-03-27

757. Measurement of differential cross sections for the production of top quark pairs and of additional jets in lepton+jets events from pp collisions at  $\sqrt{s}=13$  TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 112003](#) | [arXiv:1803.08856](#) | [DOI:10.1103/PhysRevD.97.112003](#) | 2018-03-23

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Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 789](#) | [arXiv:1803.08030](#) | [DOI:10.1140/epjc/s10052-018-6242-x](#) | 2018-03-21

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[JHEP 06 \(2018\) 101](#) | [arXiv:1803.06986](#) | [DOI:10.1007/JHEP06\(2018\)101](#) | 2018-03-19

760. Search for additional neutral MSSM Higgs bosons in the  $\tau\tau$  final state in proton-proton collisions at  $\sqrt{s}=13$  TeV

Sirunyan, Albert M et al.

[JHEP 09 \(2018\) 007](#) | [arXiv:1803.06553](#) | [DOI:10.1007/JHEP09\(2018\)007](#) | 2018-03-17

761. Search for high-mass resonances in dilepton final states in proton-proton collisions at  $\sqrt{s}=13$  TeV

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[JHEP 06 \(2018\) 120](#) | [arXiv:1803.06292](#) | [DOI:10.1007/JHEP06\(2018\)120](#) | 2018-03-16

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Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 066](#) | [arXiv:1803.05485](#) | [DOI:10.1007/JHEP08\(2018\)066](#) | 2018-03-14

763. Observation of proton-tagged, central (semi)exclusive production of high-mass lepton pairs in pp collisions at 13 TeV with the CMS-TOTEM precision proton spectrometer

Sirunyan, Albert M et al.

[JHEP 07 \(2018\) 153](#) | [arXiv:1803.04496](#) | [DOI:10.1007/JHEP07\(2018\)153](#) | 2018-03-12



764. Measurements of differential cross sections of top quark pair production as a function of kinematic event variables in proton-proton collisions at  $\sqrt{s}=13$  TeV

Sirunyan, A.M. et al.

[JHEP 06 \(2018\) 002](#) | [arXiv:1803.03991](#) | [DOI:10.1007/JHEP06\(2018\)002](#) | 2018-03-11

765. Search for a heavy resonance decaying into a Z boson and a vector boson in the  $\nu\nu q\bar{q}$  final state

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[JHEP 07 \(2018\) 075](#) | [arXiv:1803.03838](#) | [DOI:10.1007/JHEP07\(2018\)075](#) | 2018-03-10

766. Search for third-generation scalar leptoquarks decaying to a top quark and a  $\tau$  lepton at  $\sqrt{s}=13$  TeV

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[Eur.Phys.J.C 78 \(2018\) 707](#) | [arXiv:1803.02864](#) | [DOI:10.1140/epjc/s10052-018-6143-z](#) | 2018-03-07

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[JHEP 05 \(2018\) 006](#) | [arXiv:1803.00042](#) | [DOI:10.1007/JHEP05\(2018\)006](#) | 2018-02-28

768. Search for a heavy resonance decaying to a pair of vector bosons in the lepton plus merged jet final state at  $\sqrt{s}=13$  TeV

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[JHEP 05 \(2018\) 088](#) | [arXiv:1802.09407](#) | [DOI:10.1007/JHEP05\(2018\)088](#) | 2018-02-26

769. Search for narrow resonances in the b-tagged dijet mass spectrum in proton-proton collisions at  $\sqrt{s}=8$  TeV

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[Phys.Rev.Lett. 120 \(2018\) 201801](#) | [arXiv:1802.06149](#) | [DOI:10.1103/PhysRevLett.120.201801](#) | 2018-02-16

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[Phys.Rev.D 97 \(2018\) 072010](#) | [arXiv:1802.04867](#) | [DOI:10.1103/PhysRevD.97.072010](#) | 2018-02-13

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Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 221801](#) | [arXiv:1802.02965](#) | [DOI:10.1103/PhysRevLett.120.221801](#) | 2018-02-08

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Sirunyan, Albert M et al.

[JHEP 07 \(2018\) 161](#) | [arXiv:1802.02613](#) | [DOI:10.1007/JHEP07\(2018\)161](#) | 2018-02-07

773. Search for natural and split supersymmetry in proton-proton collisions at  $\sqrt{s}=13$  TeV in final states with jets and missing transverse momentum

Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 025](#) | [arXiv:1802.02110](#) | [DOI:10.1007/JHEP05\(2018\)025](#) | 2018-02-06

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[JHEP 06 \(2018\) 031](#) | [arXiv:1802.01486](#) | [DOI:10.1007/JHEP06\(2018\)031](#) | 2018-02-05

775. Search for lepton-flavor violating decays of heavy resonances and quantum black holes to  $e\mu$  final states in proton-proton collisions at  $\sqrt{s}=13$  TeV

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[JHEP 04 \(2018\) 073](#) | [arXiv:1802.01122](#) | [DOI:10.1007/JHEP04\(2018\)073](#) | 2018-02-04

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[JHEP 03 \(2018\) 181](#) | [arXiv:1802.00707](#) | [DOI:10.1007/JHEP03\(2018\)181](#) | 2018-02-02

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Sirunyan, Albert M et al.

[Phys.Rev.Lett. 121 \(2018\) 242301](#) | [arXiv:1801.04895](#) | [DOI:10.1103/PhysRevLett.121.242301](#) | 2018-01-15

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[Eur.Phys.J.C 78 \(2018\) 708](#) | [arXiv:1801.03535](#) | [DOI:10.1140/epjc/s10052-018-6146-9](#) | 2018-01-10

779. Search for new physics in events with two soft oppositely charged leptons and missing transverse momentum in proton-proton collisions at  $\sqrt{s}= 13$  TeV

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[Phys.Lett.B 782 \(2018\) 440](#) | [arXiv:1801.01846](#) | [DOI:10.1016/j.physletb.2018.05.062](#) | 2018-01-05

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[JHEP 05 \(2018\) 127](#) | [arXiv:1801.00359](#) | [DOI:10.1007/JHEP05\(2018\)127](#) | 2017-12-31

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[Eur.Phys.J.C 78 \(2018\) 589](#) | [arXiv:1712.09814](#) | [DOI:10.1140/epjc/s10052-018-6049-9](#) | 2017-12-28

782. Search for R-parity violating supersymmetry in pp collisions at  $\sqrt{s} = 13$  TeV using b jets in a final state with a single lepton, many jets, and high sum of large-radius jet masses

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[Phys.Lett.B 783 \(2018\) 114](#) | [arXiv:1712.08920](#) | [DOI:10.1016/j.physletb.2018.06.028](#) | 2017-12-24

783. Measurement of prompt and nonprompt charmonium suppression in PbPb collisions at 5.02 TeV

Sirunyan, Albert M. et al.

[Eur.Phys.J.C 78 \(2018\) 509](#) | [arXiv:1712.08959](#) | [DOI:10.1140/epjc/s10052-018-5950-6](#) | 2017-12-24

784. Search for Physics Beyond the Standard Model in Events with High-Momentum Higgs Bosons and Missing Transverse Momentum in Proton-Proton Collisions at 13 TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 241801](#) | [arXiv:1712.08501](#) | [DOI:10.1103/PhysRevLett.120.241801](#) | 2017-12-22

785. Search for lepton flavour violating decays of the Higgs boson to  $\mu\tau$  and  $e\tau$  in proton-proton collisions at  $\sqrt{s}= 13$  TeV

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[JHEP 06 \(2018\) 001](#) | [arXiv:1712.07173](#) | [DOI:10.1007/JHEP06\(2018\)001](#) | 2017-12-19

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[Phys.Rev.C 97 \(2018\) 064912](#) | [arXiv:1712.07198](#) | [DOI:10.1103/PhysRevC.97.064912](#) | 2017-12-19

787. Identification of heavy-flavour jets with the CMS detector in pp collisions at 13 TeV

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[JINST 13 \(2018\) P05011](#) | [arXiv:1712.07158](#) | [DOI:10.1088/1748-0221/13/05/P05011](#) | 2017-12-19

788. Search for the X(5568) state decaying into  $B^0_{\text{str}} \pi^\pm$  in proton-proton collisions at  $\sqrt{s} = 8 \text{ TeV}$

Sirunyan, A. M. et al.

[Phys.Rev.Lett. 120 \(2018\) 202005](#) | [arXiv:1712.06144](#) | [DOI:10.1103/PhysRevLett.120.202005](#) | 2017-12-17

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[Eur.Phys.J.C 78 \(2018\) 566](#) | [arXiv:1712.05471](#) | [DOI:10.1140/epjc/s10052-018-6033-4](#) | 2017-12-14

790. Search for  $Z\gamma$  resonances using leptonic and hadronic final states in proton-proton collisions at  $\sqrt{s} = 13 \text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 09 \(2018\) 148](#) | [arXiv:1712.03143](#) | [DOI:10.1007/JHEP09\(2018\)148](#) | 2017-12-08

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[Phys.Lett.B 779 \(2018\) 358](#) | [arXiv:1712.02825](#) | [DOI:10.1016/j.physletb.2018.02.025](#) | 2017-12-07

792. Search for the flavor-changing neutral current interactions of the top quark and the Higgs boson which decays into a pair of b quarks at  $\sqrt{s} = 13 \text{ TeV}$

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[JHEP 06 \(2018\) 102](#) | [arXiv:1712.02399](#) | [DOI:10.1007/JHEP06\(2018\)102](#) | 2017-12-06

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Sirunyan, Albert M et al.

[JHEP 02 \(2018\) 032](#) | [arXiv:1712.02280](#) | [DOI:10.1007/JHEP02\(2018\)032](#) | 2017-12-06

794. Search for new physics in final states with an energetic jet or a hadronically decaying W or Z boson and transverse momentum imbalance at  $\sqrt{s} = 13 \text{ TeV}$

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[Phys.Rev.D 97 \(2018\) 092005](#) | [arXiv:1712.02345](#) | [DOI:10.1103/PhysRevD.97.092005](#) | 2017-12-06

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[Phys.Lett.B 778 \(2018\) 349](#) | [arXiv:1711.10949](#) | [DOI:10.1016/j.physletb.2018.01.049](#) | 2017-11-29

796. Study of jet quenching with isolated-photon+jet correlations in PbPb and pp collisions at  $\sqrt{s_{\text{NN}}} = 5.02 \text{ TeV}$

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[Phys.Lett.B 785 \(2018\) 14](#) | [arXiv:1711.09738](#) | [DOI:10.1016/j.physletb.2018.07.061](#) | 2017-11-27

797. Search for new long-lived particles at  $\sqrt{s} = 13 \text{ TeV}$

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[Phys.Lett.B 780 \(2018\) 432](#) | [arXiv:1711.09120](#) | [DOI:10.1016/j.physletb.2018.03.019](#) | 2017-11-24

798. Search for gauge-mediated supersymmetry in events with at least one photon and missing transverse momentum in pp collisions at  $\sqrt{s} = 13 \text{ TeV}$

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[Phys.Lett.B 780 \(2018\) 118](#) | [arXiv:1711.08008](#) | [DOI:10.1016/j.physletb.2018.02.045](#) | 2017-11-21

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[Phys.Lett.B 789 \(2019\) 643](#) | [arXiv:1711.05594](#) | [DOI:10.1016/j.physletb.2018.11.063](#) | 2017-11-15

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[Phys.Lett.B 781 \(2018\) 390](#) | [arXiv:1711.04652](#) | [DOI:10.1016/j.physletb.2018.04.007](#) | 2017-11-13
801. Search for ZZ resonances in the  $2\ell 2\nu$  final state in proton-proton collisions at 13 TeV  
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[JHEP 03 \(2018\) 003](#) | [arXiv:1711.04370](#) | [DOI:10.1007/JHEP03\(2018\)003](#) | 2017-11-12
802. Measurement of the underlying event activity in inclusive Z boson production in proton-proton collisions at  $\sqrt{s}=13\text{ TeV}$   
 Sirunyan, A.M. et al.  
[JHEP 07 \(2018\) 032](#) | [arXiv:1711.04299](#) | [DOI:10.1007/JHEP07\(2018\)032](#) | 2017-11-12
803. Measurement of the inclusive tt cross section in pp collisions at  $\sqrt{s}=5.02\text{ TeV}$  using final states with at least one charged lepton  
 Sirunyan, A.M. et al.  
[JHEP 03 \(2018\) 115](#) | [arXiv:1711.03143](#) | [DOI:10.1007/JHEP03\(2018\)115](#) | 2017-11-08
804. Measurement of the cross section for top quark pair production in association with a W or Z boson in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$   
 Sirunyan, Albert M et al.  
[JHEP 08 \(2018\) 011](#) | [arXiv:1711.02547](#) | [DOI:10.1007/JHEP08\(2018\)011](#) | 2017-11-07
805. Measurement of associated Z + charm production in proton-proton collisions at  $\sqrt{s} = 8\text{ TeV}$   
 Sirunyan, A.M. et al.  
[Eur.Phys.J.C 78 \(2018\) 287](#) | [arXiv:1711.02143](#) | [DOI:10.1140/epjc/s10052-018-5752-x](#) | 2017-11-06
806. Search for top squarks and dark matter particles in opposite-charge dilepton final states at  $\sqrt{s}= 13\text{ TeV}$   
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[Phys.Rev.D 97 \(2018\) 032009](#) | [arXiv:1711.00752](#) | [DOI:10.1103/PhysRevD.97.032009](#) | 2017-11-02
807. Search for new physics in events with a leptonically decaying Z boson and a large transverse momentum imbalance in proton-proton collisions at  $\sqrt{s} = 13\text{ TeV}$   
 Sirunyan, A.M. et al.  
[Eur.Phys.J.C 78 \(2018\) 291](#) | [arXiv:1711.00431](#) | [DOI:10.1140/epjc/s10052-018-5740-1](#) | 2017-11-01
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[Phys.Lett.B 780 \(2018\) 251](#) | [arXiv:1710.11002](#) | [DOI:10.1016/j.physletb.2018.02.033](#) | 2017-10-30
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[Phys.Rev.D 97 \(2018\) 012007](#) | [arXiv:1710.11188](#) | [DOI:10.1103/PhysRevD.97.012007](#) | 2017-10-30
810. Search for supersymmetry in events with at least three electrons or muons, jets, and missing transverse momentum in proton-proton collisions at  $\sqrt{s}=13\text{ TeV}$   
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[JHEP 02 \(2018\) 067](#) | [arXiv:1710.09154](#) | [DOI:10.1007/JHEP02\(2018\)067](#) | 2017-10-25
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812. Measurement of b hadron lifetimes in pp collisions at  $\sqrt{s} = 8$  TeV

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[Eur.Phys.J.C 78 \(2018\) 457](#) | [arXiv:1710.08949](#) | [DOI:10.1140/epjc/s10052-018-5929-3](#) | 2017-10-24

813. Measurement of differential cross sections in the kinematic angular variable  $\phi^*$  for inclusive Z boson production in pp collisions at  $\sqrt{s} = 8$  TeV

Sirunyan, A.M. et al.

[JHEP 03 \(2018\) 172](#) | [arXiv:1710.07955](#) | [DOI:10.1007/JHEP03\(2018\)172](#) | 2017-10-22

814. Pseudorapidity and transverse momentum dependence of flow harmonics in pPb and PbPb collisions

Sirunyan, Albert M et al.

[Phys.Rev.C 98 \(2018\) 044902](#) | [arXiv:1710.07864](#) | [DOI:10.1103/PhysRevC.98.044902](#) | 2017-10-21

815. Search for a massive resonance decaying to a pair of Higgs bosons in the four b quark final state in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, A.M. et al.

[Phys.Lett.B 781 \(2018\) 244](#) | [arXiv:1710.04960](#) | [DOI:10.1016/j.physletb.2018.03.084](#) | 2017-10-13

816. Measurement of angular parameters from the decay  $B^0 \rightarrow K^{*0} \mu^+ \mu^-$  in proton-proton collisions at  $\sqrt{s} = 8$  TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 781 \(2018\) 517](#) | [arXiv:1710.02846](#) | [DOI:10.1016/j.physletb.2018.04.030](#) | 2017-10-08

817. Study of dijet events with a large rapidity gap between the two leading jets in pp collisions at  $\sqrt{s} = 7$  TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 242](#) | [arXiv:1710.02586](#) | [DOI:10.1140/epjc/s10052-018-5691-6](#) | 2017-10-06

818. Search for pair production of vector-like quarks in the bWbW channel from proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 779 \(2018\) 82](#) | [arXiv:1710.01539](#) | [DOI:10.1016/j.physletb.2018.01.077](#) | 2017-10-04

819. Search for low mass vector resonances decaying into quark-antiquark pairs in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[JHEP 01 \(2018\) 097](#) | [arXiv:1710.00159](#) | [DOI:10.1007/JHEP01\(2018\)097](#) | 2017-09-30

820. Search for supersymmetry in events with one lepton and multiple jets exploiting the angular correlation between the lepton and the missing transverse momentum in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, A.M. et al.

[Phys.Lett.B 780 \(2018\) 384](#) | [arXiv:1709.09814](#) | [DOI:10.1016/j.physletb.2018.03.028](#) | 2017-09-28

821. Observation of Correlated Azimuthal Anisotropy Fourier Harmonics in pp and p+Pb Collisions at the LHC

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 092301](#) | [arXiv:1709.09189](#) | [DOI:10.1103/PhysRevLett.120.092301](#) | 2017-09-26

822. Measurements of the  $pp \rightarrow ZZ$  production cross section and the  $Z \rightarrow 4\ell$  branching fraction, and constraints on anomalous triple gauge couplings at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 165](#) | [arXiv:1709.08601](#) | [DOI:10.1140/epjc/s10052-018-5567-9](#) | 2017-09-25

823. Observation of top quark production in proton-nucleus collisions

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 119 \(2017\) 242001](#) | [arXiv:1709.07411](#) | [DOI:10.1103/PhysRevLett.119.242001](#) | 2017-09-21

824. Evidence for the Higgs boson decay to a bottom quark-antiquark pair

Sirunyan, Albert M et al.

[Phys.Lett.B 780 \(2018\) 501](#) | [arXiv:1709.07497](#) | [DOI:10.1016/j.physletb.2018.02.050](#) | 2017-09-21

825. Observation of electroweak production of same-sign W boson pairs in the two jet and two same-sign lepton final state in proton-proton collisions at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 081801](#) | [arXiv:1709.05822](#) | [DOI:10.1103/PhysRevLett.120.081801](#) | 2017-09-18

826. Inclusive search for a highly boosted Higgs boson decaying to a bottom quark-antiquark pair

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[Phys.Rev.Lett. 120 \(2018\) 071802](#) | [arXiv:1709.05543](#) | [DOI:10.1103/PhysRevLett.120.071802](#) | 2017-09-16

827. Combination of inclusive and differential tt charge asymmetry measurements using ATLAS and CMS data at  $\sqrt{s}=7$  and 8 TeV

Aaboud, Morad et al.

[JHEP 04 \(2018\) 033](#) | [arXiv:1709.05327](#) | [DOI:10.1007/JHEP04\(2018\)033](#) | 2017-09-15

828. Search for electroweak production of charginos and neutralinos in multilepton final states in proton-proton collisions at  $\sqrt{s}= 13$  TeV

Sirunyan, A.M. et al.

[JHEP 03 \(2018\) 166](#) | [arXiv:1709.05406](#) | [DOI:10.1007/JHEP03\(2018\)166](#) | 2017-09-15

829. Search for Higgsino pair production in pp collisions at  $\sqrt{s} = 13$  TeV in final states with large missing transverse momentum and two Higgs bosons decaying via  $H \rightarrow b \bar{b}$

Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 032007](#) | [arXiv:1709.04896](#) | [DOI:10.1103/PhysRevD.97.032007](#) | 2017-09-14

830. Search for supersymmetry with Higgs boson to diphoton decays using the razor variables at  $\sqrt{s} = 13$  TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 779 \(2018\) 166](#) | [arXiv:1709.00384](#) | [DOI:10.1016/j.physletb.2017.12.069](#) | 2017-09-01

831. Measurement of the Splitting Function in pp and Pb-Pb Collisions at  $\sqrt{s_{NN}} = 5.02$  TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 142302](#) | [arXiv:1708.09429](#) | [DOI:10.1103/PhysRevLett.120.142302](#) | 2017-08-30

832. Probing the chiral magnetic wave in pPb and PbPb collisions at  $\sqrt{s_{NN}} = 5.02$  TeV using charge-dependent azimuthal anisotropies

Sirunyan, Albert M et al.

[Phys.Rev.C 100 \(2019\) 064908](#) | [arXiv:1708.08901](#) | [DOI:10.1103/PhysRevC.100.064908](#) | 2017-08-29

833. Search for heavy resonances decaying to a top quark and a bottom quark in the lepton+jets final state in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 777 \(2018\) 39](#) | [arXiv:1708.08539](#) | [DOI:10.1016/j.physletb.2017.12.006](#) | 2017-08-28

834. Search for Evidence of the Type-III Seesaw Mechanism in Multilepton Final States in Proton-Proton Collisions at  $\sqrt{s}=13$  TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 119 \(2017\) 221802](#) | [arXiv:1708.07962](#) | [DOI:10.1103/PhysRevLett.119.221802](#) | 2017-08-26

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Sirunyan, A.M. et al.

[JHEP 04 \(2018\) 060](#) | [arXiv:1708.07638](#) | [DOI:10.1007/JHEP04\(2018\)060](#) | 2017-08-25

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[Phys.Rev.C 96 \(2017\) 064902](#) | [arXiv:1708.07113](#) | [DOI:10.1103/PhysRevC.96.064902](#) | 2017-08-23

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Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 072006](#) | [arXiv:1708.05379](#) | [DOI:10.1103/PhysRevD.97.072006](#) | 2017-08-17

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Sirunyan, Albert M et al.

[Phys.Lett.B 782 \(2018\) 474](#) | [arXiv:1708.04962](#) | [DOI:10.1016/j.physletb.2018.05.074](#) | 2017-08-16

839. Search for resonant and nonresonant Higgs boson pair production in the  $b\bar{b}\nu\ell\nu$  final state in proton-proton collisions at  $\sqrt{s}=13\text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 01 \(2018\) 054](#) | [arXiv:1708.04188](#) | [DOI:10.1007/JHEP01\(2018\)054](#) | 2017-08-14

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Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 202301](#) | [arXiv:1708.03497](#) | [DOI:10.1103/PhysRevLett.120.202301](#) | 2017-08-11

841. Measurement of vector boson scattering and constraints on anomalous quartic couplings from events with four leptons and two jets in proton-proton collisions at  $\sqrt{s}=13\text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 774 \(2017\) 682](#) | [arXiv:1708.02812](#) | [DOI:10.1016/j.physletb.2017.10.020](#) | 2017-08-09

842. Search for vectorlike light-flavor quark partners in proton-proton collisions at  $\sqrt{s}=8\text{ TeV}$

Sirunyan, A. M. et al.

[Phys.Rev.D 97 \(2018\) 072008](#) | [arXiv:1708.02510](#) | [DOI:10.1103/PhysRevD.97.072008](#) | 2017-08-08

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[Phys.Rev.C 97 \(2018\) 044912](#) | [arXiv:1708.01602](#) | [DOI:10.1103/PhysRevC.97.044912](#) | 2017-08-04

844. Search for single production of a vector-like T quark decaying to a Z boson and a top quark in proton-proton collisions at  $\sqrt{s}=13\text{ TeV}$

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[Phys.Lett.B 781 \(2018\) 574](#) | [arXiv:1708.01062](#) | [DOI:10.1016/j.physletb.2018.04.036](#) | 2017-08-03

845. Observation of the Higgs boson decay to a pair of  $\tau$  leptons with the CMS detector

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[Phys.Lett.B 779 \(2018\) 283](#) | [arXiv:1708.00373](#) | [DOI:10.1016/j.physletb.2018.02.004](#) | 2017-08-01

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[JHEP 11 \(2017\) 010](#) | [arXiv:1707.07283](#) | [DOI:10.1007/JHEP11\(2017\)010](#) | 2017-07-23



847. Search for the pair production of third-generation squarks with two-body decays to a bottom or charm quark and a neutralino in proton-proton collisions at  $\sqrt{s} = 13$  TeV

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[Phys.Lett.B 778 \(2018\) 263](#) | [arXiv:1707.07274](#) | [DOI:10.1016/j.physletb.2018.01.012](#) | 2017-07-23

848. Measurement of the differential cross sections for the associated production of a W boson and jets in proton-proton collisions at  $\sqrt{s}=13$  TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 96 \(2017\) 072005](#) | [arXiv:1707.05979](#) | [DOI:10.1103/PhysRevD.96.072005](#) | 2017-07-19

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Sirunyan, Albert M et al.

[JHEP 12 \(2017\) 142](#) | [arXiv:1707.06193](#) | [DOI:10.1007/JHEP12\(2017\)142](#) | 2017-07-19

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Sirunyan, Albert M et al.

[JHEP 10 \(2017\) 005](#) | [arXiv:1707.03316](#) | [DOI:10.1007/JHEP10\(2017\)005](#) | 2017-07-11

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Sirunyan, A.M. et al.

[JHEP 03 \(2018\) 167](#) | [arXiv:1707.03325](#) | [DOI:10.1007/JHEP03\(2018\)167](#) | 2017-07-11

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[Phys.Lett.B 778 \(2018\) 101](#) | [arXiv:1707.02909](#) | [DOI:10.1016/j.physletb.2018.01.001](#) | 2017-07-10

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[Eur.Phys.J.C 77 \(2017\) 636](#) | [arXiv:1707.01303](#) | [DOI:10.1140/epjc/s10052-017-5192-z](#) | 2017-07-05

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[Phys.Lett.B 775 \(2017\) 1](#) | [arXiv:1707.00541](#) | [DOI:10.1016/j.physletb.2017.10.021](#) | 2017-07-03

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[Phys.Rev.D 96 \(2017\) 072004](#) | [arXiv:1707.00350](#) | [DOI:10.1103/PhysRevD.96.072004](#) | 2017-07-02

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Sirunyan, Albert M et al.

[Phys.Rev.D 96 \(2017\) 112003](#) | [arXiv:1706.10194](#) | [DOI:10.1103/PhysRevD.96.112003](#) | 2017-06-30

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[DOI:10.17181/CERN.QZ28.FLHW](#) | 2017-06-30

858. Measurements of properties of the Higgs boson decaying into the four-lepton final state in pp collisions at  $\sqrt{s}=13$  TeV

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[JHEP 11 \(2017\) 047](#) | [arXiv:1706.09936](#) | [DOI:10.1007/JHEP11\(2017\)047](#) | 2017-06-29

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Khachatryan, Vardan et al.

[arXiv:1706.08310](#) | 2017-06-26

860. Search for a heavy composite Majorana neutrino in the final state with two leptons and two quarks at  $\sqrt{s}=13$  TeV

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[Phys.Lett.B 775 \(2017\) 315](#) | [arXiv:1706.08578](#) | [DOI:10.1016/j.physletb.2017.11.001](#) | 2017-06-26

861. Measurement of the semileptonic  $t\bar{t} + \gamma$  production cross section in pp collisions at  $\sqrt{s}=8$  TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2017\) 006](#) | [arXiv:1706.08128](#) | [DOI:10.1007/JHEP10\(2017\)006](#) | 2017-06-25

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Sirunyan, Albert M et al.

[JHEP 10 \(2017\) 131](#) | [arXiv:1706.05868](#) | [DOI:10.1007/JHEP10\(2017\)131](#) | 2017-06-19

863. Suppression of Excited  $\Upsilon$  States Relative to the Ground State in Pb-Pb Collisions at  $\sqrt{s_{NN}}=5.02$  TeV

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[Phys.Rev.Lett. 120 \(2018\) 142301](#) | [arXiv:1706.05984](#) | [DOI:10.1103/PhysRevLett.120.142301](#) | 2017-06-19

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[JINST 12 \(2017\) P10003](#) | [arXiv:1706.04965](#) | [DOI:10.1088/1748-0221/12/10/P10003](#) | 2017-06-15

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Sirunyan, Albert M et al.

[JHEP 08 \(2017\) 029](#) | [arXiv:1706.04260](#) | [DOI:10.1007/JHEP08\(2017\)029](#) | 2017-06-13

866. Search for new physics in the monophoton final state in proton-proton collisions at  $\sqrt{s}=13$  TeV

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[JHEP 10 \(2017\) 073](#) | [arXiv:1706.03794](#) | [DOI:10.1007/JHEP10\(2017\)073](#) | 2017-06-12

867. Search for pair production of vector-like T and B quarks in single-lepton final states using boosted jet substructure in proton-proton collisions at  $\sqrt{s}=13$  TeV

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[JHEP 11 \(2017\) 085](#) | [arXiv:1706.03408](#) | [DOI:10.1007/JHEP11\(2017\)085](#) | 2017-06-11

868. Search for dark matter produced in association with heavy-flavor quark pairs in proton-proton collisions at  $\sqrt{s}=13$  TeV

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[Eur.Phys.J.C 77 \(2017\) 845](#) | [arXiv:1706.02581](#) | [DOI:10.1140/epjc/s10052-017-5317-4](#) | 2017-06-08

869. Search for top quark partners with charge  $5/3$  in proton-proton collisions at  $\sqrt{s}=13$  TeV

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[JHEP 08 \(2017\) 073](#) | [arXiv:1705.10967](#) | [DOI:10.1007/JHEP08\(2017\)073](#) | 2017-05-31

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[Phys.Rev.Lett. 119 \(2017\) 111802](#) | [arXiv:1705.10532](#) | [DOI:10.1103/PhysRevLett.119.111802](#) | 2017-05-30

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Sirunyan, Albert M et al.

[Phys.Lett.B 776 \(2018\) 355](#) | [arXiv:1705.10141](#) | [DOI:10.1016/j.physletb.2017.11.043](#) | 2017-05-29

872. Combination of searches for heavy resonances decaying to WW, WZ, ZZ, WH, and ZH boson pairs in proton–proton collisions at  $\sqrt{s}=8$  and 13 TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 774 \(2017\) 533](#) | [arXiv:1705.09171](#) | [DOI:10.1016/j.physletb.2017.09.083](#) | 2017-05-25

873. Search for Supersymmetry in pp Collisions at  $\sqrt{s}=13$ TeV in the Single-Lepton Final State Using the Sum of Masses of Large-Radius Jets

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[Phys.Rev.Lett. 119 \(2017\) 151802](#) | [arXiv:1705.04673](#) | [DOI:10.1103/PhysRevLett.119.151802](#) | 2017-05-12

874. Measurement of the  $B^{\pm}$  Meson Nuclear Modification Factor in Pb-Pb Collisions at  $\sqrt{s_{NN}}=5.02$ TeV

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[Phys.Rev.Lett. 119 \(2017\) 152301](#) | [arXiv:1705.04727](#) | [DOI:10.1103/PhysRevLett.119.152301](#) | 2017-05-12

875. Search for Charged Higgs Bosons Produced via Vector Boson Fusion and Decaying into a Pair of W and Z Bosons Using pp Collisions at  $\sqrt{s}=13$ TeV

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[Phys.Rev.Lett. 119 \(2017\) 141802](#) | [arXiv:1705.02942](#) | [DOI:10.1103/PhysRevLett.119.141802](#) | 2017-05-08

876. Measurement of the triple-differential dijet cross section in proton-proton collisions at  $\sqrt{s}=8$  TeV and constraints on parton distribution functions

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[Eur.Phys.J.C 77 \(2017\) 746](#) | [arXiv:1705.02628](#) | [DOI:10.1140/epjc/s10052-017-5286-7](#) | 2017-05-07

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[Phys.Lett.B 774 \(2017\) 279](#) | [arXiv:1705.01403](#) | [DOI:10.1016/j.physletb.2017.09.053](#) | 2017-05-03

878. Search for supersymmetry in multijet events with missing transverse momentum in proton-proton collisions at 13 TeV

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[Phys.Rev.D 96 \(2017\) 032003](#) | [arXiv:1704.07781](#) | [DOI:10.1103/PhysRevD.96.032003](#) | 2017-04-25

879. Measurement of the top quark mass in the dileptonic  $t\bar{t}$  decay channel using the mass observables  $M_{b\ell}$ ,  $M_{T2}$ , and  $M_{b\ell\nu}$  in pp collisions at  $\sqrt{s}=8$  TeV

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[Phys.Rev.D 96 \(2017\) 032002](#) | [arXiv:1704.06142](#) | [DOI:10.1103/PhysRevD.96.032002](#) | 2017-04-20

880. Search for  $t\bar{t}$  resonances in highly boosted lepton+jets and fully hadronic final states in proton-proton collisions at  $\sqrt{s}=13$  TeV

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[JHEP 07 \(2017\) 001](#) | [arXiv:1704.03366](#) | [DOI:10.1007/JHEP07\(2017\)001](#) | 2017-04-11

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Sirunyan, A.M. et al.

[JHEP 10 \(2017\) 072](#) | [arXiv:1704.00366](#) | [DOI:10.1007/JHEP10\(2017\)072](#) | 2017-04-02

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[JHEP 07 \(2017\) 013](#) | [arXiv:1703.09986](#) | [DOI:10.1007/JHEP07\(2017\)013](#) | 2017-03-29

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[Eur.Phys.J.C 77 \(2017\) 467](#) | [arXiv:1703.06330](#) | [DOI:10.1140/epjc/s10052-017-5030-3](#) | 2017-03-18

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[Phys.Lett.B 772 \(2017\) 21](#) | [arXiv:1703.06095](#) | [DOI:10.1016/j.physletb.2017.06.009](#) | 2017-03-17

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887. Search for third-generation scalar leptoquarks and heavy right-handed neutrinos in final states with two tau leptons and two jets in proton-proton collisions at  $\sqrt{s}=13$  TeV

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[JHEP 07 \(2017\) 121](#) | [arXiv:1703.03995](#) | [DOI:10.1007/JHEP07\(2017\)121](#) | 2017-03-11

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[Eur.Phys.J.C 77 \(2017\) 354](#) | [arXiv:1703.02530](#) | [DOI:10.1140/epjc/s10052-017-4912-8](#) | 2017-03-07

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[Eur.Phys.J.C 77 \(2017\) 459](#) | [arXiv:1703.01630](#) | [DOI:10.1140/epjc/s10052-017-4984-5](#) | 2017-03-05

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[JHEP 07 \(2017\) 014](#) | [arXiv:1703.01651](#) | [DOI:10.1007/JHEP07\(2017\)014](#) | 2017-03-05

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[Phys.Lett.B 772 \(2017\) 336](#) | [arXiv:1702.06164](#) | [DOI:10.1016/j.physletb.2017.06.064](#) | 2017-02-20

892. Measurement of the cross section for electroweak production of  $Z\gamma$  in association with two jets and constraints on anomalous quartic gauge couplings in proton-proton collisions at  $\sqrt{s} = 8$  TeV

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[Phys.Lett.B 770 \(2017\) 380](#) | [arXiv:1702.03025](#) | [DOI:10.1016/j.physletb.2017.04.071](#) | 2017-02-09

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[Eur.Phys.J.C 77 \(2017\) 269](#) | [arXiv:1702.01462](#) | [DOI:10.1140/epjc/s10052-017-4828-3](#) | 2017-02-05

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901. Measurement of the  $t\bar{t}$  production cross section using events with one lepton and at least one jet in pp collisions at  $\sqrt{s} = 13$  TeV

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[JHEP 09 \(2017\) 051](#) | [arXiv:1701.06228](#) | [DOI:10.1007/JHEP09\(2017\)051](#) | 2017-01-22

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Sirunyan, Albert M et al.

[JINST 12 \(2017\) P04023](#) | [arXiv:1701.02022](#) | [DOI:10.1088/1748-0221/12/04/P04023](#) | 2017-01-08

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[Phys.Lett.B 770 \(2017\) 278](#) | [arXiv:1612.09274](#) | [DOI:10.1016/j.physletb.2017.04.043](#) | 2016-12-29

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[JHEP 03 \(2017\) 162](#) | [arXiv:1612.09159](#) | [DOI:10.1007/JHEP03\(2017\)162](#) | 2016-12-29

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[JHEP 03 \(2017\) 077](#) | [arXiv:1612.01190](#) | [DOI:10.1007/JHEP03\(2017\)077](#) | 2016-12-04

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[JHEP 03 \(2017\) 101](#) | [arXiv:1611.08931](#) | [DOI:10.1007/JHEP03\(2017\)101](#) | 2016-11-27

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[Eur.Phys.J.C 77 \(2017\) 751](#) | [arXiv:1611.06507](#) | [DOI:10.1140/epjc/s10052-017-5140-y](#) | 2016-11-20

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[Phys.Lett.B 769 \(2017\) 391](#) | [arXiv:1611.06604](#) | [DOI:10.1016/j.physletb.2017.04.005](#) | 2016-11-20

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[JHEP 02 \(2017\) 048](#) | [arXiv:1611.06594](#) | [DOI:10.1007/JHEP02\(2017\)048](#) | 2016-11-20

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[Phys.Lett.B 769 \(2017\) 520](#) | [arXiv:1611.03568](#) | [DOI:10.1016/j.physletb.2017.02.012](#) | 2016-11-10

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Khachatryan, Vardan et al.

[Phys.Lett.B 770 \(2017\) 357](#) | [arXiv:1611.01510](#) | [DOI:10.1016/j.physletb.2017.04.031](#) | 2016-11-04

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[Phys.Rev.Lett. 118 \(2017\) 162301](#) | [arXiv:1611.01438](#) | [DOI:10.1103/PhysRevLett.118.162301](#) | 2016-11-04

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[Eur.Phys.J.C 77 \(2017\) 294](#) | [arXiv:1611.00338](#) | [DOI:10.1140/epjc/s10052-017-4787-8](#) | 2016-11-01

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[Phys.Lett.B 770 \(2017\) 50](#) | [arXiv:1610.09551](#) | [DOI:10.1016/j.physletb.2017.04.028](#) | 2016-10-29

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[JHEP 05 \(2017\) 013](#) | [arXiv:1610.07095](#) | [DOI:10.1007/JHEP05\(2017\)013](#) | 2016-10-22

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[Phys.Rev.D 95 \(2017\) 012009](#) | [arXiv:1610.05133](#) | [DOI:10.1103/PhysRevD.95.012009](#) | 2016-10-17

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Khachatryan, Vardan et al.

[JHEP 04 \(2017\) 018](#) | [arXiv:1610.04870](#) | [DOI:10.1007/JHEP04\(2017\)018](#) | 2016-10-16

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Khachatryan, Vardan et al.

[JHEP 02 \(2017\) 079](#) | [arXiv:1610.04857](#) | [DOI:10.1007/JHEP02\(2017\)079](#) | 2016-10-16

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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

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[Phys.Lett.B 768 \(2017\) 57](#) | [arXiv:1609.05391](#) | [DOI:10.1016/j.physletb.2017.02.010](#) | 2016-09-17

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 Khachatryan, Vardan et al.  
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 Khachatryan, Vardan et al.  
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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

[JHEP 02 \(2016\) 156](#) | [arXiv:1601.00079](#) | [DOI:10.1007/JHEP02\(2016\)156](#) | 2016-01-01

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[Phys.Lett.B 759 \(2016\) 36](#) | [arXiv:1512.06461](#) | [DOI:10.1016/j.physletb.2016.05.044](#) | 2015-12-20

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[Phys.Rev.Lett. 116 \(2016\) 071801](#) | [arXiv:1512.01224](#) | [DOI:10.1103/PhysRevLett.116.071801](#) | 2015-12-03

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[Eur.Phys.J.C 76 \(2016\) 155](#) | [arXiv:1512.00815](#) | [DOI:10.1140/epjc/s10052-016-3988-x](#) | 2015-12-02

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[Phys.Lett.B 758 \(2016\) 321](#) | [arXiv:1511.06170](#) | [DOI:10.1016/j.physletb.2016.05.005](#) | 2015-11-19

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Khachatryan, Vardan et al.

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[Phys.Rev.D 93 \(2016\) 032004](#) | [arXiv:1509.03744](#) | [DOI:10.1103/PhysRevD.93.032004](#) | 2015-09-12

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[JHEP 11 \(2015\) 018](#) | [arXiv:1508.07774](#) | [DOI:10.1007/JHEP11\(2015\)018](#) | 2015-08-31

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[Eur.Phys.J.C 76 \(2016\) 13](#) | [arXiv:1508.07819](#) | [DOI:10.1140/epjc/s10052-015-3853-3](#) | 2015-08-31

1041. Search for supersymmetry in the vector-boson fusion topology in proton-proton collisions at  $\sqrt{s}=8$  TeV

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[JHEP 11 \(2015\) 189](#) | [arXiv:1508.07628](#) | [DOI:10.1007/JHEP11\(2015\)189](#) | 2015-08-30

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[Phys.Rev.Lett. 116 \(2016\) 032301](#) | [arXiv:1508.06678](#) | [DOI:10.1103/PhysRevLett.116.032301](#) | 2015-08-26

1043. Search for  $W'$  decaying to tau lepton and neutrino in proton-proton collisions at  $\sqrt{s} = 8$  TeV

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[Phys.Lett.B 755 \(2016\) 196](#) | [arXiv:1508.04308](#) | [DOI:10.1016/j.physletb.2016.02.002](#) | 2015-08-18

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[Phys.Rev.D 93 \(2016\) 034014](#) | [arXiv:1508.03862](#) | [DOI:10.1103/PhysRevD.93.034014](#) | 2015-08-16

1045. Search for neutral MSSM Higgs bosons decaying to  $\mu^+ \mu^-$  in pp collisions at  $\sqrt{s} = 7$  and 8 TeV

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[Phys.Lett.B 752 \(2016\) 221](#) | [arXiv:1508.01437](#) | [DOI:10.1016/j.physletb.2015.11.042](#) | 2015-08-06

1046. Search for supersymmetry in events with a photon, a lepton, and missing transverse momentum in pp collisions at  $\sqrt{s}= 8$  TeV

Khachatryan, Vardan et al.

- Phys.Lett.B 757 (2016) 6 | [arXiv:1508.01218](#) | [DOI:10.1016/j.physletb.2016.03.039](#) | 2015-08-05
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[Phys.Lett.B 753 \(2016\) 424](#) | [arXiv:1507.08126](#) | [DOI:10.1016/j.physletb.2015.12.020](#) | 2015-07-29
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Khachatryan, Vardan et al.

[Phys.Lett.B 743 \(2015\) 383](#) | [arXiv:1409.5761](#) | [DOI:10.1016/j.physletb.2015.02.048](#) | 2014-09-19

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Khachatryan, Vardan et al.

[Phys.Rev.Lett. 114 \(2015\) 061801](#) | [arXiv:1409.4789](#) | [DOI:10.1103/PhysRevLett.114.061801](#) | 2014-09-16

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Khachatryan, Vardan et al.

[Phys.Lett.B 742 \(2015\) 200](#) | [arXiv:1409.3392](#) | [DOI:10.1016/j.physletb.2015.01.034](#) | 2014-09-11

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Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 235](#) | [arXiv:1408.3583](#) | [DOI:10.1140/epjc/s10052-015-3451-4](#) | 2014-08-15

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Khachatryan, Vardan et al.

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[Phys.Rev.D 91 \(2015\) 052008](#) | [arXiv:1408.3104](#) | [DOI:10.1103/PhysRevD.91.052008](#) | 2014-08-13

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Khachatryan, Vardan et al.

[JHEP 09 \(2014\) 087](#) | [arXiv:1408.1682](#) | [DOI:10.1007/JHEP09\(2014\)087](#) | 2014-08-07

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Khachatryan, Vardan et al.

[Phys.Lett.B 739 \(2014\) 229](#) | [arXiv:1408.0806](#) | [DOI:10.1016/j.physletb.2014.10.063](#) | 2014-08-04

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Khachatryan, Vardan et al.

[Phys.Lett.B 739 \(2014\) 23](#) | [arXiv:1407.6643](#) | [DOI:10.1016/j.physletb.2014.10.032](#) | 2014-07-24

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[Eur.Phys.J.C 74 \(2014\) 3149](#) | [arXiv:1407.3683](#) | [DOI:10.1140/epjc/s10052-014-3149-z](#) | 2014-07-14

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Khachatryan, Vardan et al.

[Phys.Lett.B 740 \(2015\) 83](#) | [arXiv:1407.3476](#) | [DOI:10.1016/j.physletb.2014.11.026](#) | 2014-07-13

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Khachatryan, Vardan et al.

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[Phys.Lett.B 740 \(2015\) 250](#) | [arXiv:1406.0113](#) | [DOI:10.1016/j.physletb.2016.04.010](#) | 2014-05-31

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[Eur.Phys.J.C 74 \(2014\) 3129](#) | [arXiv:1405.7225](#) | [DOI:10.1140/epjc/s10052-014-3129-3](#) | 2014-05-28

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[Phys.Rev.D 90 \(2014\) 112001](#) | [arXiv:1405.3961](#) | [DOI:10.1103/PhysRevD.90.112001](#) | 2014-05-15

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[Phys.Lett.B 736 \(2014\) 371](#) | [arXiv:1405.3886](#) | [DOI:10.1016/j.physletb.2014.07.053](#) | 2014-05-15

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[Phys.Lett.B 736 \(2014\) 64](#) | [arXiv:1405.3455](#) | [DOI:10.1016/j.physletb.2014.06.077](#) | 2014-05-14

1153. Search for Massive Resonances in Dijet Systems Containing Jets Tagged as W or Z Boson Decays in pp Collisions at  $\sqrt{s} = 8$  TeV

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[Phys.Rev.D 90 \(2014\) 032008](#) | [arXiv:1404.4619](#) | [DOI:10.1103/PhysRevD.90.032008](#) | 2014-04-17

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[Eur.Phys.J.C 74 \(2015\) 3014](#) | [arXiv:1404.3171](#) | [DOI:10.1140/epjc/s10052-014-3014-0](#) | 2014-04-11

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[Phys.Lett.B 736 \(2014\) 33](#) | [arXiv:1404.2292](#) | [DOI:10.1016/j.physletb.2014.06.076](#) | 2014-04-08

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[Eur.Phys.J.C 74 \(2014\) 2980](#) | [arXiv:1404.1344](#) | [DOI:10.1140/epjc/s10052-014-2980-6](#) | 2014-04-04

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[Phys.Lett.B 730 \(2014\) 243](#) | [arXiv:1310.0878](#) | [DOI:10.1016/j.physletb.2014.01.042](#) | 2013-10-02

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Chatrchyan, Serguei et al.

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1239. Measurement of the  $t\bar{t}$  Production Cross Section in the  $\tau$  +Jets Channel in pp Collisions at  $\sqrt{s} = 7$  TeV

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